

When Women Run Against Men: Evidence from Political Platforms

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Abstract

Do female and male candidates campaign differently, and why? Using individual political platforms from French legislative elections between 1981 and 2024, I combine a transformer machine learning model with a regression discontinuity design that exploits the two-round structure of French elections to identify the causal impact of candidate gender on campaign content. Female and male candidates do not emphasize different issues in the first-round; differences emerge only once the campaign narrows to two candidates. Between the two rounds, women’s relative emphasis on male-stereotyped topics rises by 5 percentage points, driven by male candidates, who withdraw from these topics and reallocate space to framing the national contest. Women’s relative emphasis on female-stereotyped topics falls by 2.2 points, driven by female candidates reducing their emphasis on health. These adjustments are concentrated among candidates who trail after the first-round, yet they bring no detectable electoral return. I provide suggestive evidence that women reduce the prevalence of female-stereotyped topics only when they compete against a woman, and using an LLM-based measure of agentic and communal language, that they adopt a more communal register only when they compete against a man, consistent with the behavior gender stereotypes prescribe.

Keywords: Elections, gender, natural language processing

JEL classification: D72, J16, P0

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1 Introduction

The share of women in politics has increased remarkably over the past few decades in almost every country. In 1988, women accounted for only 8.9% of candidates in US primaries; by 2024, they accounted for 26.2% (the highest share was in 2020, 29.1%).¹ In France, the share increased from 11.5% to 41.1% in the same period. The rising number of women in politics might significantly affect the policies discussed if women and men have different preferences. How do women differ from men during political campaigns, and why do such differences arise? Understanding these differences is crucial because candidate communication shapes the information available to voters and can influence which policy areas gain political attention.

To secure election, candidates must navigate the art of campaign communication and carefully decide which policy issues to advertise and how to communicate them. How candidates choose to communicate may differ between men and women. Voters associate men with competence on security, crime, and defense, and women with competence on health, education, and compassion issues (Huddy & Terkildsen, 1993). Role congruity theory holds that deviating from the behavior expected of one's gender is evaluated more harshly for women than for men (Eagly & Karau, 2002; Rudman & Glick, 2001). Stereotypes operate not only through how others evaluate a person but also through beliefs about one's own ability: Coffman (2014) finds that individuals are less willing to contribute their ideas in domains perceived as incongruent with their gender, even conditional on actual ability, and Bordalo, Coffman, Gennaioli, and Shleifer (2019) show that these beliefs are shaped by the stereotype-relevant features of the comparison at hand. If candidates hold such beliefs, or expect voters to hold them, then broadening a campaign is not symmetric: a man who moves onto security issues and a woman who moves onto health are not making the same strategic bet, even when the electoral logic pushing them to move is identical.

Studying how female and male candidates differ in their campaigns is methodologically challenging. Female candidacy is endogenous to local electoral conditions, and systematic data on campaign discourse are scarce. In most contexts, records of candidates' campaign platforms are either incomplete or limited to those who ultimately win office, leading the literature to focus primarily on behavior once elected.

I overcome these challenges by focusing on French legislative elections, which use a two-round system in which the same electorate votes twice. With an average

¹Source: Center for American Women and Politics.

of nine candidates in the first-round and two in the second, candidates face incentives to adjust their messages between rounds, creating a natural setting to observe strategic signaling in response to voter evaluations. To address endogeneity, I exploit quasi-experimental variation in the candidate's gender. Using a regression discontinuity design, I examine whether female (male) candidates shift their platform topics between the first and second-rounds. The research design compares a female politician who barely passes to the second-round instead of a male in that district with a male politician who barely qualifies instead of a female. To support the identification strategy, I show that a large set of voters' and districts' characteristics, as well as the candidate's individual characteristics (except gender), are balanced at the threshold.

The empirical analysis draws on a comprehensive dataset of candidates' official campaign platforms, which the French state distributes to all registered voters. Using a transformer-based machine learning model built specifically to understand and process the French language (CamemBERT), I quantify topic prevalence. I find that in the first-round, female and male candidates do not differ significantly in how they run their campaigns. Between the first and second-round, the gap on male-stereotyped topics (justice, criminality and immigration) increases by 5.0 percentage points. At the same time, women decrease the prevalence of female-stereotyped topics (health and education), especially health, by 2.4 percentage points. Who moves? Decomposing the estimate into the average change on each side of the threshold, I conclude that each gender reduces the prevalence of the respective stereotype topic between rounds. Men who narrowly qualify reduce the salience of male-stereotyped topics by 3.9 percentage points, while women decrease the prevalence of female-stereotyped topics by 3.2 percentage points.

Several experiments outside politics have shown that women's performance in competitive environments depends on their opponents' gender (e.g., Gneezy, Niederle, and Rustichini (2003), Gneezy and Rustichini (2004), Iriberry and Rey-Biel (2017)). I provide suggestive evidence that the opponent's gender matters in competitive environments, but only for females. Women decrease the prevalence of female-stereotyped topics only when they compete against a woman. The prevalence of male-stereotyped topics does not change depending on the opponent. Men broaden their campaigns away from their stereotype regardless of whom they face; women do so only when their opponent is also a woman, that is, when gender stereotypes cannot distinguish them from their rival.

Against a man, women's adjustment takes a different form: it concerns how they speak rather than what they speak about. Using a large language model, I classify each segment of the platforms as agentic, presenting the candidate as

determined, strong, and in control, or communal, presenting the candidate as oriented toward others, and build an index that takes higher values when the language is relatively more agentic and lower values when it is relatively more communal. Between the first and second-round, women increase their use of communal language when they compete against a man, and do not adjust it when they compete against a woman. This pattern is consistent with role congruity theory, but only in mixed-gender races, where gender is most salient to voters.

These adjustments respond to competitive pressure. They are concentrated among candidates who enter the runoff behind their opponent: each additional point of first-round disadvantage widens the gender gap in health and male-stereotyped topics, while among front-runners, male and female platforms are adjusted similarly.

Are candidates' strategic behavior electorally compensated? Following Acharya, Blackwell, and Sen (2016), I estimate average controlled direct effects by sequential g-estimation to ask whether the adjustment affects second-round vote shares. It does not. No detectable gender gap emerges in runoff performance, and fixing the adjustment at the same value for all candidates leaves that gap unchanged; the estimated return to adjustment is small and statistically indistinguishable from zero for both men and women. That candidates adjust along gendered lines even though the adjustment has no detectable electoral return is more consistent with shared expectations about how men and women are supposed to campaign than with a vote-maximizing response to voter preferences.

Contribution to the literature The paper contributes to three main strands of the literature. First, it contributes to the literature on gender and political representation. Several studies have shown that women prefer to work on policies that benefit female voters (Chattopadhyay and Duflo (2004)) or in domains such as health and child care, which are usually stereotyped as female (Bhalotra and Clots-Figueras (2014), Funk and Gathmann (2015), Lippmann (2021), Baskaran and Hessami (2025)). In contrast, other studies find no significant differences (Ferreira and Gyourko (2014), Bagues and Campa (2021), Casarico, Lattanzio, and Profeta (2022)). This literature is concentrated on elected politicians, as it is considerably easier to obtain data. To my knowledge, my paper provides, for the first time, a causal description of gender differences during political campaigns. The results show that the gender composition of a race affects the issues discussed, a margin of political representation that operates before any woman is elected.

Second, it contributes to the literature on gender stereotypes and behavior. Experimental work in economics shows that stereotypes operate through beliefs about one's own ability in gender-incongruent domains (Coffman, 2014) and that

such beliefs respond to the composition of the relevant comparison (Bordalo et al., 2019), while work on competitive environments shows that behavior depends on the gender composition of the contest (Gneezy et al., 2003; Iriberry & Rey-Biel, 2017; Niederle & Vesterlund, 2007). This pattern has been confirmed in elections (Kanthak and Woon (2015), Barber, Butler, Preece, et al. (2016)). These works have largely focused on entry decisions and laboratory settings. I study the behavior among candidates who have already entered and reached the final stage of a high-stakes competition in a quasi-experimental setting. I find that in competitive environments, female candidates adapt to their opponent’s gender, conforming to what is stereotypically expected.

Third, it contributes to the literature on strategic platform choice. The Median Voter Theorem (Downs (1957)) assumes the parties’ unique objective is winning elections. Consequently, if two parties have the same information about voters’ preferences, they will converge on the median voter’s viewpoint, ignoring politicians’ identities. However, candidates only converge to a certain extent. Citizen-candidate models highlight that politicians also care about policy implementation, not only victory (Alesina (1988), Osborne and Slivinski (1996), Besley and Coate (1997)). Le Pennec (2023) shows that French candidates adjust the content of their communication between rounds as the electorate they address widens, and Di Tella, Kotti, Le Pennec, and Pons (2025) document strategic platform adjustment in French and U.S. elections. I show that these adjustments are gendered and that the gender composition of the runoff changes the campaign’s overall issue content.

2 Institutional context

2.1 French parliamentary elections

The paper focuses on parliamentary elections that elect all members of the National Assembly, the lower house of Parliament.

These elections are held under direct universal suffrage with a two-round plurality voting rule. Candidates can win in the first-round if they receive more than 50 percent of the votes and at least 25 percent of the registered voters. However, in 86.62% of elections, no candidate is elected in the first round, leading to a second-round held one week later. The second-round is decided by simple plurality: the candidate with the largest vote share wins.

Candidates who obtain at least 12.5 percent of the eligible voters’ vote share qualify for the second round. However, if only one candidate (or none) meets the threshold, the two candidates with the largest vote share can proceed to the second

round. This last situation accounts for 26.93% of all elections.

Selection of political candidates It is possible to run for one of the 577 French constituencies without being affiliated to a party as long as the aspiring MP meets all the necessary conditions to enter the race, in particular being at least 18 years old, having the right to vote, and not being ineligible (because of a court decision or a function incompatible with the mandate of an MP, such as being mayor). However, most candidates run under a party label, making them much more visible to voters.

Under the 2000 gender parity law, parties must present an equal fraction of male and female candidates across the electoral districts. If the difference between female and male candidates exceeds 4% (48% females and 52% males, or the reverse), non-compliance with the gender parity rules results in a financial penalty. The financial penalty is calculated as follows: "public funding provided to political parties based on the number of votes they receive in the first-round of elections is reduced by a percentage equivalent to one-half of the difference between the total number of candidates of each sex, out of the total number of candidates" (Le Barbanchon and Sauvagnat (2022)). In France, voters vote for two candidates: the leading candidate, who, if elected, becomes an MP, and the substitute. The last one replaces the top candidate only if he leaves office; potential reasons include being nominated to a function incompatible with the mandate of an MP (e.g., minister, secretary of state, or elected as a mayor), or death/health issues. In most cases, the substitute never becomes an MP during the legislative term and only has a symbolic role. After the law's approval, in the 2002 legislative elections, women accounted for 38.8% of the leading candidates, compared to 23% in the 1997 legislative elections.

2.2 Political platforms

During the legislative campaign, individual candidates can emit one political platform (*trans.* profession de foi) before each election round.² The appendix provides one example in Figure B.1. What is the traditional content of a political platform? A manifesto allows a candidate to present the program and ideas they plan to commit to if elected. Since they represent an electoral district, candidates might run a campaign by focusing on national policies and/or addressing local issues. They can also run a more partisan program, personalize their campaign, focus on preferred policy topics, or, as expected, appeal to voters and criticize the opposition.

²Throughout the paper, I use the words "political platform" and "manifesto" interchangeably; in this paper, they are synonymous.

Candidates must print these platforms, and the state can reimburse the costs if they win at least 5% of the votes in one of the rounds (Electoral law, articles R39 and L216). An official local propaganda committee is responsible for mailing the manifestos to voters at least four days before the first round and three days before the second round (if it happens).³

According to the Ipsos - CEVIPOF 2022 Presidential electoral survey, the primary sources of candidates' information are: 38% of the electors follow television, 15% the internet (many candidates share their platforms online), 13% newspapers, 12% the manifestos received in the mailbox, 6% use other sources around them, and 16% do not inform. Relatively to the 1988, 1993, and 1997 manifestos, the internet was not yet widely available and used. Furthermore, per election, there are 577 constituencies and an average of 5800 candidates, making it impossible for television to cover them all. Consequently, platforms are likely more important in legislative than presidential elections.

3 Data

3.1 Electoral data

Each dataset records the number of registered voters, abstentions, cast votes, valid and invalid votes, and each candidate's votes in each electoral district. From 1981 to 2017, I use data from Granzier, Pons, and Tricaud (2023). The dataset also includes information on the candidate's first and last name, gender, political affiliation, incumbency status, and whether the candidate ran in the past. For 2022 and 2024, I use data from the Ministry of Interior.

The Ministry of Interior has also provided information on candidates' dates of birth and occupations since 2012. Several candidates provide personal information in their manifestos, including their occupation and age. Before 2012, I prompted ChatGPT to read each political program and provide the main candidate's occupation and age. I then construct five dummy variables: out of the labor force (e.g., unemployed, retired, student), high-skilled worker (e.g., managers, engineers, lawyers), middle-skilled (e.g., agents, social assistants, merchants), low-skilled (e.g., craftsmen, factory workers, farmers), and a missing category.

I use the official party labels provided by the Ministry of the Interior to classify candidates (including independent ones), and following Jolly et al. (2022), I

³These platforms are only mailed if they obey these rules: they must have a maximum size of 210x297 millimeters, weigh between 60 and 80 grams per square meter (Electoral law, article R29) (Le Pennec (2023)).

classify them into six partisan families: far-left, left, liberal, right, far-right, and other. The last category includes politicians who do not fit into any of these traditional ideological categories or who do not identify with any ideology. In the rest of the paper, I refer to political orientation as the broader categories "left" (far-left and left) and "right" (liberal, right and far-right), unless specified otherwise. Essentially, I classify "left" parties as the ones that score less than 5 in the left-right dimension of Jolly et al. (2022) and "right" as the ones that score more than 6.⁴ Party classification is available in Appendix Section A.

Campaign contributions Data on campaign expenditures and contributions from 1993 to 2017 come from Bekkouche, Cagé, and Dewitte (2022). For 2022 and 2024, I use data from the National Commission on Campaign Accounts and Political Financing (CNCCFP). In French parliamentary elections, candidates who receive at least 1% of the votes in the first-round must submit their campaign accounts to the French CNCCFP. This commission was created in 1990, so data before that date is unavailable. For each year, electoral district, and candidate, I observe, for the whole campaign, the total amount spent by the candidate, the total amount of contributions he received, and the amount of each type of contribution: contributions received from the candidate's political party, donations, and personal funds.

3.2 Political platforms

The Archelec project digitized candidate manifestos for the 1981, 1988 and 1993 elections (Gaultier-Voituriez (2016)).⁵⁶ Platforms for the 1997 elections were digitized from the National Archives. For the 2002, 2007 and 2012 elections, I use data from Di Tella et al. (2025).⁷ The data for 2012 are complemented with data from the French National Assembly website, which provides the political platforms of all elected politicians.⁸

⁴According to Jolly et al. (2022), all liberal parties in the sample have a left-right score between 6 and 7, in a spectrum between 0 and 10.

⁵They are available at <https://archive.org/details/archiveselectoralesducevipof/>

⁶The Archelec project (Gaultier-Voituriez (2016)) includes data since 1958; however, I only start in 1988, since before this period any observation would be included in the regression discontinuity design; there are no women who barely passed to the second round.

⁷Di Tella et al. (2025) have comprehensively collected data for the 2021 local elections. However, each *canton* elects two councilors: a male and a female candidate who campaign on a common platform; therefore, it is not possible to study gender differences in local political campaigns, and the article focuses solely on parliamentary campaigns.

⁸They are available at: <https://www.assemblee-nationale.fr/dyn/decouvrir-1-assemblee/histoire/barodet2/recueil-des-professions-de-foi-de-la-14eme-legislature>.

For 2017, they come from several sources: during the campaign, the Ministry of Interior shares the manifestos submitted by the candidates on their website⁹, and they were web-scraped by a non-profit organization called RegardsCitoyens¹⁰. In addition, I manually digitized missing manifestos at CEVIPOF (Sciences Po), the National Assembly website also shares the manifestos of all elected MPs¹¹, some missing manifestos were also found on several local news or candidate websites. For 2022 and 2024, platforms come from the Ministry of Interior website and several local news and candidate websites.

The dataset comprises 33686 political platforms, 26630 for the first-round, and 7056 for the second-round. Optical character recognition (Mistral OCR) transformed these platforms from images to text.

3.3 Measurement

3.3.1 Topics

To measure the thematic content of political text, I classify each sentence in the corpus into one or two of eleven topics using a supervised multi-label classifier based on CamemBERT (Antoun et al. (2024)), a transformer language model pre-trained on French text. The classification is performed at the paragraph level, and the corpus comprises approximately 597 thousand sentences. The classification proceeds in three stages: (i) construction of an annotated training set using a large language model (LLM), (ii) fine-tuning of CamemBERT on this training set, and (iii) inference on the full corpus.

I adopt a multi-label framework: each sentence may be assigned to more than one topic simultaneously. This choice reflects the nature of political discourse, where a single sentence often spans several policy domains—for example, a statement about the green economy belongs to both the economic and environmental domains.

I define eleven mutually intelligible categories: (1) Economy, (2) Employment and social security, (3) Environment, (4) Health, (5) Education, (6) Justice and criminality, (7) Immigration, (8) Foreign policy, (9) Local politics, (10) National politics, and (11) Other, a residual category reserved for procedural language, fragments, and sentences without identifiable content.

⁹They are available at: <https://programme-candidats.interieur.gouv.fr/>

¹⁰They are available at: <https://github.com/regardscitoyens/professions-foi-candidats/tree/master/documents/LG17>

¹¹They are available at: <https://www.assemblee-nationale.fr/dyn/decouvrir-1-assemblee/histoire/barodet2/recueil-des-professions-de-foi-de-la-15eme-legislature>

Supervised classification requires annotated training data. I annotated about 50,000 sentences sampled from both manifesto corpora using GPT-5.4-mini and a structured French-language prompt. In Appendix Table B2, I share the prompt provided, translated into English.

I apply the fine-tuned classifier to all 597 thousand paragraphs. For each sentence, the model outputs a probability for each of the nine topics; assign a topic when its probability exceeds 0.6, and if no topic clears the threshold, assign the sentence 'other'. For each paragraph, I retain the full set of assigned topics, the highest-probability (top) topic, the second-ranked topic, and the associated probabilities.

To reduce measurement error and facilitate interpretation, I aggregate all topics into three main categories: (1) female-stereotyped topics: health and education; (2) male-stereotyped topics: justice, immigration, and foreign policy; (3) other: the remaining topics. The empirical analysis aggregates paragraph-level classifications to the candidate level and computes each topic's share in the manifesto/debates.

Validation Transformers are a powerful approach for topic classification because they can capture the context and semantic meaning of words within a sentence, rather than relying primarily on word frequencies or predefined dictionaries. When fine-tuned for topic classification, CamemBERT can therefore identify nuanced thematic content in French political texts and distinguish between closely related topics. Since the topic measure is central to every result that follows, I assess its quality along three dimensions: internal classification performance, agreement with human judgment, and agreement with an independent automated classifier.

Table B3 reports per-class precision, recall, and F1-scores across the eleven topic categories. Performance is uniformly high and, importantly, does not deteriorate for the categories that carry the analysis: health ($F1 = 0.88$), education (0.86), justice and criminality (0.87), and immigration (0.82). The lowest scores appear for immigration and the residual category, which is expected given their smaller training support and, in the case of the residual category, its definition as a catch-all rather than a substantive topic.

The analysis was also validated against an independent human annotator. A random sample of 1000 paragraphs was provided to the human annotator, and the classification was done into female-stereotyped topics (health and education), male-stereotyped topics (justice & criminality and immigration), and other. Table B4 reports the results. The classifier reaches an F1-score of 0.746 for female-stereotyped content, 0.764 for male-stereotyped content, and 0.957 for the residual category. Recall exceeds precision for female-stereotyped paragraphs (0.824 ver-

sus 0.682), and the pattern reverses for male-stereotyped paragraphs (0.748 versus 0.782), indicating mild over-assignment of the former and mild under-assignment of the latter relative to human judgment.

In addition, the analysis was also validated against a large language model (ChatGPT 5.4 version) and, finally, between the LLM and the human annotator. Tables B5 and B6 report the comparisons, respectively. The language model agrees with the human annotator at almost the same rate as the classifier: F1-scores of 0.763 and 0.754 for the two stereotyped categories, versus 0.746 and 0.764 for the classifier.

3.3.2 Emotion & reason

To capture gender differences in how candidates discuss political issues, I implement the emotion–reason measure developed in Gennaro and Ash (2022) and adapt it to French political texts. The measure locates each document on a continuous scale between two poles: appeals grounded in affect, emotion, and moral sentiment on one end, and appeals grounded in cognition, evidence, and analytical reasoning on the other. Whereas the topic classifier captures what politicians talk about, this measure captures how they talk about it.

The construction follows Gennaro and Ash (2022) in three steps. First, I train word embeddings on the corpus using Word2Vec (Mikolov, Sutskever, Chen, Corrado, and Dean (2013)). Second, I define an emotion pole and a reason pole using validated seed dictionaries: the Affective Processing and Cognitive Processing categories of the French edition of LIWC (Piolat, Booth, Chung, Davids, and Pennebaker (2011)). Each pole is represented as the average embedding of its seed words, expanded to include each seed’s ten nearest neighbors in the trained embedding space, so the generic dictionary adapts to the vocabulary of French political discourse. Third, each document is represented as the average of its word vectors, and its emotionality score is computed as the difference between its cosine similarity to the emotion pole and its cosine similarity to the reason pole.

Validation In the Appendix, I show the validation exercises performed. Table B7 shows the most emotional and rational words, respectively.

3.3.3 Agentic vs communal language

I build on the distinction between agency and communal, the two fundamental dimensions of social judgment (Abele & Wojciszke, 2007). Agentic language presents the candidate as acting with determination, strength, and control: fighting for an outcome, deciding, leading, and obtaining results. Communal language presents

the candidate as oriented toward others: listening, being close to and available for constituents, attending to people’s concerns, and acting in cooperation with them. Role congruity theory (Eagly & Karau, 2002) predicts that these registers carry different costs and benefits for women and men seeking office, since the political role is stereotypically agentic while women are expected to be communal.

A central feature of the measure is that it captures the register in which candidates present themselves, not the topics they address. I classify each segment with a large language model (ChatGPT 5.4-mini version). I use two separate binary prompts, one for each register, so that a segment can be agentic, communal, both, or neither. Treating the two dimensions independently follows the psychological literature, in which agency and communal are distinct dimensions rather than opposite poles of a single scale. In Appendix Tables B8 and B9, I share the prompt provided for agentic and communal language, respectively.

I prefer this approach to dictionary methods for two reasons. First, dictionaries cannot identify who performs an action (e.g., “we will ask the government to remain attentive to the elderly”, the communal word refers to the government, not the candidate). Second, it is not easy to build a dictionary that captures all agentic and communal words, and some may be conflated with policy vocabulary (e.g., solidarity, support, energy).

The *agentic-communal* measure equals the difference between the standardized share of sentences classified as agentic and the standardized share of sentences classified as communal.

4 Empirical strategy

Do female candidates campaign differently from male candidates? Answering this question is empirically challenging because female presence in a race is endogenous. Previous research has shown that various factors such as party discrimination (Esteve-Volart and Bagues (2012), Lippmann (2021)), voter bias (Le Barbanchon and Sauvagnat (2022)), or previous female success might affect women’s willingness to run (Beaman, Chattopadhyay, Duflo, Pande, and Topalova (2009), Bhavnani (2009), Baskaran and Hessami (2018)). A quasi-experiment is necessary to understand how gender affects politicians’ strategic behavior in a competitive electoral environment.

As noted in Section 2.1, in races where only one or no candidate received 12.5% of the votes, only the two most-voted candidates are eligible for the second round. I focus on races in which the second-place candidate’s vote share was below 12.5%. To estimate the causal impact of gender on campaign strategies, I use a sharp

regression discontinuity design (RDD) and compare female candidates who narrowly passed to the second round (while a male was not barely eligible in that electoral district) with male candidates who narrowly passed to the second round (while a female was almost eligible). I estimate the following equation:

$$Y_i = \beta_0 + \beta_1 T_i + \beta_2 R_i + \beta_3 R_i T_i + \mu_i \quad (1)$$

The outcome variable Y_i is the difference in the share of a policy topic between the second and the first round. The treatment variable T is a dummy equal to one if the candidate is a woman and 0 if the candidate is a man in the second round. The running variable R is the difference between the share of votes obtained by the most-voted female and male candidate, ignoring the opponent in the second round. The unit of observation is the candidate.

Following Calonico, Cattaneo, Farrell, and Titiunik (2019), I nonparametrically estimate the coefficient of interest using local linear regressions. I use the mean squared error optimal bandwidths (MSERD) selection procedure proposed by Calonico, Cattaneo, and Titiunik (2014). This procedure is data-driven, implying that bandwidth size varies with the outcome under consideration.

Sample In Table C10, I provide summary statistics comparing all candidates with those included in the sample and within a cutoff of 3 percentage points. Table C11 shows the number of observations included in the sampling frame each year. Most of the variation available in the data comes from the elections held in 2017, 2022, and 1993. Figure C.3 shows that the districts included in the sample - where a female or male candidate was within 3 percentage points of the cutoff - span the French territory, and neither males nor females are concentrated in a specific geographic area.

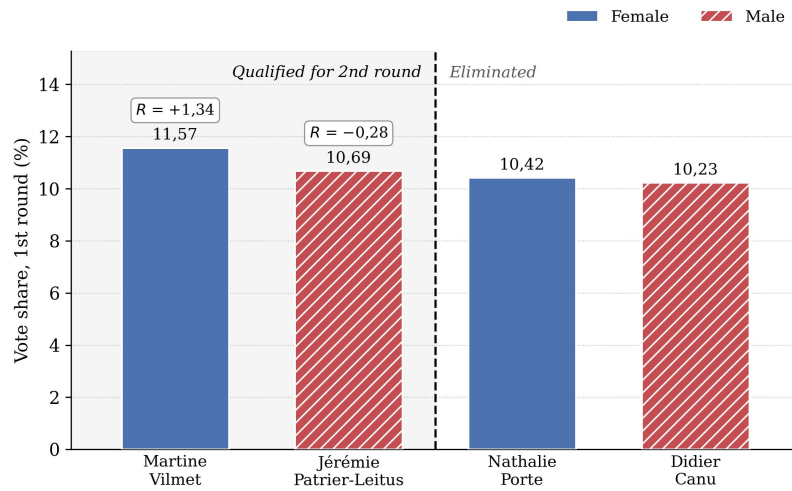
4.1 Internal validity

The validity of the RDD relies on the assumption that first-round candidates of a particular type (e.g., males) do not systematically sort on the right of the qualification threshold. Such manipulation is unlikely since France is a democracy, and international observers qualify the elections as pluralistic, competitive, and respectful of fundamental rights.¹² In addition, manipulation is difficult because it requires predicting the outcome of the first election stage with great accuracy. I test the assumption's validity using the Cattaneo, Jansson, and Ma (2018) test and check whether the density of the running variable jumps at the threshold. As

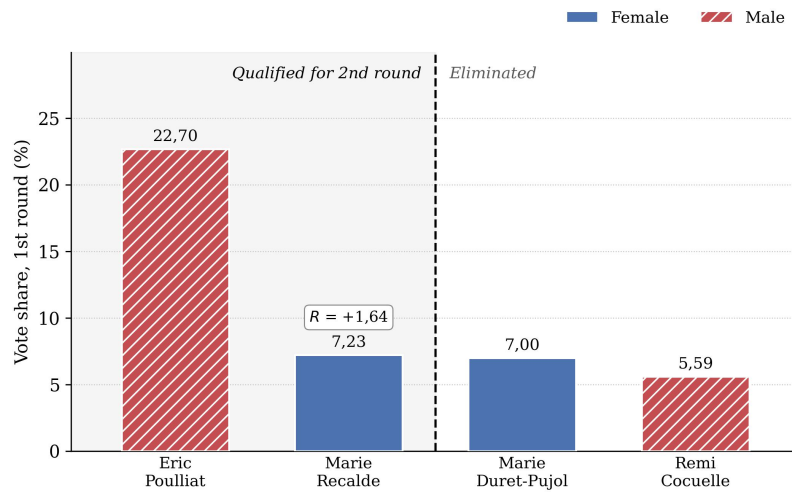
¹²France. *Presidential Election 10 and 24 April 2022. ODIHR Election Assessment Mission Final Report* (2022).

Figure 1: Examples of running variable construction

(a) Department 14 district 3, election-year 2022



(b) Department 33 district 6, election-year 2017



Notes. The two examples show the electoral results for the four most-voted candidates in two different electoral districts. In the case of Panel (a), two observations are included in the analysis, while in the case of Panel (b), only one observation is included.

Figure C.4 in the Appendix demonstrates, there is no jump at the margin.

Another implication of the identifying assumption is that districts' characteristics are continuous at the threshold. I run balancing tests for turnout, number of candidates, number of female candidates, number of enrolled voters, whether a woman was elected in the past, gender wage gap, number of candidates per ideology, and sum of vote share per ideology. Table C12 in the Appendix shows sharp RDD estimates of the effect of having a woman in the second-round, rather than a man, on first-round electoral district characteristics. Of the 18 regressions, the coefficient on the treatment variable is statistically significant only for the number

of enrolled voters at the 5 percent. Reassuringly, the main results are similar when controlling for this covariate (Appendix Table D20).

I also test for a jump in several individual characteristics. Marshall (2024) notes that politician regression discontinuity designs identify the effect of the predetermined characteristic and all compensating differentials. I run balancing tests for several individual characteristics (Appendix Table C13): vote share in the first-round, victory margin between the candidate and the opponent in the second-round, incumbent, run in the past, occupation (out of labor force, low-skilled, middle-skilled, high-skilled), age groups, and ideology (far-left, left, liberal, right, far-right, other). Only one out of the 20 individual characteristics is statistically significant: right, at 10 percent. As a robustness check, I run a separate regression discontinuity analysis in which I focus on male candidates and compare how right-wing vs non-right-wing candidates adjust between rounds; I do not find significant differences in the adjustment of female- and male-stereotyped topics, and the adjustment is close to zero (Appendix Table C15). I also run a heterogeneity analysis in Table C16 and conclude that right-wing politicians drive the results.

Di Tella et al. (2025) show empirically that candidates strategically adjust their platforms to move closer to their opponents. The results may be confounded because female candidates' opponents are mostly from a specific ideology. In Table C14, I present balancing tests on opponents' ideology and find that they are balanced.

An interesting question is whether women and men already differed in first-round topic salience or campaigning style. Tables 1 show no significant differences, since the values are close to zero and nonsignificant. Table C17 shows that candidates also do not differ significantly in campaigning style, meaning in the use of emotional or rhetorical language when addressing political issues.

Next, I run a general balance test to check whether being a female candidate versus a male candidate jumps at the threshold. To predict treatment status, I regress treatment status on all the covariates referred to above and in Tables C12 and C13. As shown in Appendix Table C18 and Figure C.5, the predicted probability of being a female candidate does not jump at the qualification threshold.

Table 1: Differences in topics salience between female and male candidates in the first round

	(1) Female-stereotyped	(2) Male-stereotyped	(3) Other
Woman	0.012 (0.022)	0.005 (0.035)	-0.002 (0.020)
Observations	963	963	963
Eff. number of obs	328	434	462
Robust p-value	0.689	0.810	0.951
Polyn. order	1	1	1
Bandwidth	2.724	3.719	3.946
Mean, left of threshold	0.122	0.114	0.893

Standard errors in parentheses are clustered at the district-year level. Each column reports the results from a separate local polynomial regression. The variable of interest is being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. The dependent variables take values between 0 and 1, and their computation is explained in Section 3.3.1. The mean gives the average outcome value for male candidates at the threshold.

Endogenous sample selection can also affect the analysis. Although some candidates are eligible for the second-round, they decide not to run; this constitutes only 0.8% of cases, and column 1 of Table C19 shows that female politicians are no more likely than males to give up on going to the second round. Another concern is whether one gender is more likely to have a political platform observed than the other. Column 2 of Table C19 shows that this is not the case; there is no significant jump in the probability of having both manifestos available at the qualification threshold.

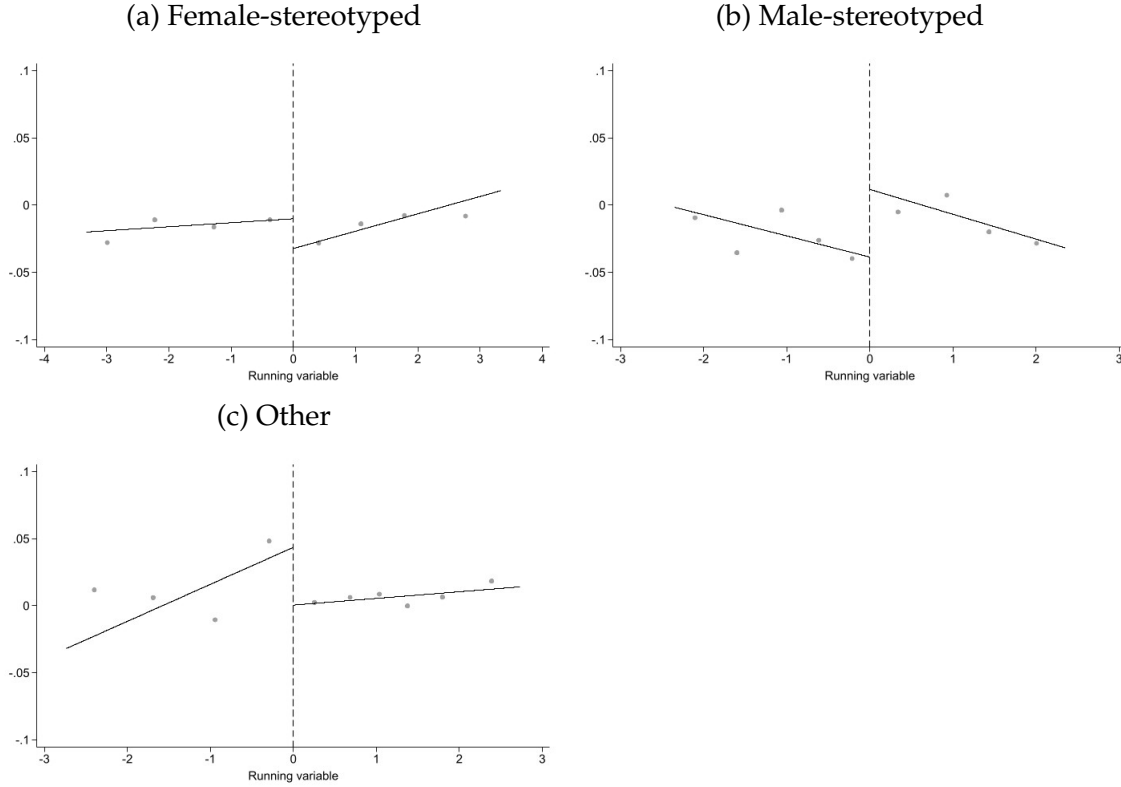
5 Results: Gender differences in political platforms

5.1 Topics

In this section, I use the empirical strategy described above to examine gender differences in how politicians adjust topic salience between the first and second-rounds. I examine the adjustment in three ways: female-stereotyped topics (health and education), male-stereotyped topics (justice and immigration), and other topics (the remaining topics).

Figure 2 shows the largest jump in male-stereotyped topics. For the remaining topics, coverage appears to drop slightly. In the figure, each dot represents the average outcome within a given bin of the running variable - that is, the vote share difference between the most-voted woman and the most-voted man in the first-round (excluding the opponent).

Figure 2: Gender differences in topics adjustment between 1st and 2nd round



Notes. Each graph is constructed by restricting the support to observations within the estimation bandwidths and fitting a local polynomial point estimator (polynomial order 1 and triangular kernel). Dots represent the local averages of the outcome (vertical axis). Averages are calculated within quantile bins of the running variable (horizontal axis). The outcome is the change in topic coverage from the first to the second-round. The running variable is the vote share difference between the most-voted woman and the most-voted man, excluding the opponent in the second round.

Table 2 complements the graphical analysis and provides the formal estimates of the effects. Panel A aggregates the eleven topics into three groups: female-stereotyped topics (health and education), male-stereotyped topics (justice and criminality, and immigration), and the remaining topics. Panel B reports the individual topics that compose the three groups. As Panel A shows, between the first and second-round, women write 2.2 percentage points less about female-stereotyped topics (not statistically significant) and 5 percentage points more about male-stereotyped topics than men (statistically significant at 5 percent). Candi-

dates have a two-page limit for their platforms, and as a consequence this increase might come at the cost of less coverage of other topics. This is indeed what column (3) shows: women decrease the coverage of other topics by 4.3 percentage points. Panel B shows how these aggregates decompose. Within male-stereotyped topics, the increase is concentrated in immigration (5.4 percentage points, robust $p = 0.065$), while the estimates for justice are smaller and not statistically significant. Within female-stereotyped topics, the reduction is concentrated in health (-2.4 percentage points, robust $p = 0.026$), while the change in education is near zero.

Table 2: Gender differences in topics adjustment between 1st and 2nd round

Panel A: Aggregated topic groups

	(1) Female-stereotyped	(2) Male-stereotyped	(3) Other
Woman	-0.022 (0.016)	0.050 (0.023)	-0.043 (0.023)
Robust p -value	0.179	0.030	0.044
Observations	963	963	963
Eff. number of obs	405	294	329
Polyn. order	1	1	1
Bandwidth	3.331	2.349	2.732
Mean, left of threshold	-0.010	-0.039	0.044

Panel B: Individual topics

	(1) Health	(2) Education	(3) Justice	(4) Immigration	(5) Local	(6) National	(7) Other policy
Woman	-0.024 (0.011)	0.002 (0.014)	0.016 (0.013)	0.054 (0.032)	0.010 (0.018)	-0.056 (0.047)	0.034 (0.049)
Robust p -value	0.026	0.718	0.308	0.065	0.713	0.210	0.382
Observations	963	963	963	963	963	963	963
Eff. number of obs	301	299	399	299	457	390	355
Polyn. order	1	1	1	1	1	1	1
Bandwidth	2.471	2.423	3.288	2.423	3.919	3.221	2.955
Mean, left of threshold	0.003	-0.018	-0.011	-0.045	-0.005	0.111	-0.054

Notes: Standard errors in parentheses are clustered at the district-year level. Each column reports the results from a separate local polynomial regression. The variable of interest is being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. Coefficients are conventional local polynomial estimates; standard errors, confidence intervals, and p -values are robust bias-corrected following Calonico et al. (2014). The dependent variables take values between 0 and 1, and their computation is explained in Section 3.3.1. The mean gives the average outcome value for male candidates at the threshold.

Robustness To further support the identification strategy, I conduct several robustness checks. The results are similar when I test the covariate-adjusted version (Calonico et al. (2019)); see Appendix Table D21. I also show the robustness of the results to a wide range of bandwidths (Appendix Table D.6). I also present

jackknife estimates excluding one candidate at a time to ensure that the results are not driven by a subset of observations near the cutoff and show that the point estimates remain very similar (Appendix Table D.7).

5.2 Campaigning style: Emotion vs reason

Studies on the role of emotions in individual-level decision-making have shown that emotional appeals can be an effective persuasive tool for politicians to influence how voters process information and respond to their messages (Bakker, Schumacher, and Rooduijn (2021), Osnabrügge, Hobolt, and Rodon (2021)). Warm and expressive candidates are expected to handle female-stereotyped topics better, while masculine instrumental traits are better perceived in male-stereotyped topics (Huddy and Terkildsen (1993)). On the other hand, female candidates have stronger incentives to employ reason, because voters already associate women with warmth and emotion and may penalize women for appearing insufficiently competent. At the same time, a woman who is more emotionally expressive in everyday life may still produce essentially the same amount of emotional campaign rhetoric as a male politician because both try to satisfy the same expectations of political professionalism.

In Table 3, I show no significant gender differences in how candidates adjust their campaigns between the first and second-round. Column (2) shows that women use more emotion when discussing male-stereotyped topics; however, this result is significant only at the 10 percent level and is not robust to other bandwidths.

6 Mechanisms

6.1 Who is adjusting? Female or male candidates?

In Section 5.1, the regression discontinuity estimates identify the difference in the adjustment between rounds between female and male candidates. The results may be driven by women increasing the prevalence of a topic, men decreasing it, or both. To address this question, I estimate the average change in topic salience at the cutoff separately for men and women who barely qualified for the second-round, using local linear regressions with a triangular kernel and the same MSE-optimal bandwidth on both sides of the threshold. Table 4 reports the results.

Male candidates drive the gender gap in male-stereotyped topics. Between the first and second-rounds, men who barely qualified reduce the salience of male-

Table 3: Differences between how female and male candidates adjust their campaigns between the 1st and 2nd round in terms of emotion vs reason

	(1) All topics	(2) Female-stereotyped	(3) Male-stereotyped	(4) Other
Woman	0.083 (0.178)	0.267 (0.151)	0.357 (0.198)	-0.163 (0.186)
Observations	963	963	963	963
Eff. number of obs	291	472	299	259
Robust p-value	0.844	0.101	0.083	0.309
Polyn. order	1	1	1	1
Bandwidth	2.317	4.024	2.427	2.063
Mean, left of threshold	-0.071	0.026	-0.088	0.083

Standard errors in parenthesis are clustered at the district-candidate level. Each column reports the results from a separate local polynomial regression. The variable of interest is a being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. The computation of the dependent variables is explained in Section 3.3.2. Higher values mean that the candidate implements more emotional language, while lower values mean that the candidate uses for more reason in his writing. The mean gives the average outcome value for male candidates at the threshold.

stereotyped topics by 3.9 percentage points ($p < 0.05$), whereas women leave the salience of these topics essentially unchanged (+1.2 percentage points, not statistically significant). The resulting 5.0 percentage-point difference is therefore explained by men moving away from these topics rather than by women moving toward them. The salience men lose appears to be reallocated to topics that are not gender-stereotyped: men increase the salience of the remaining topics by 4.4 percentage points ($p < 0.05$), while women’s emphasis on these topics does not change (0.1 percentage points). The gender gap in the remaining topics (−4.3 percentage points, $p < 0.05$) thus mirrors the gap in male-stereotyped topics.

The pattern for female-stereotyped topics differs. Women who barely qualified reduce the salience of these topics by 3.2 percentage points ($p < 0.01$), while the change for men is small and not statistically significant (−1.0 percentage points). However, the difference between women and men (−2.2 percentage points) is not statistically significant, so we cannot conclude that women retreat from female-stereotyped topics more than men do.

Panel B decomposes each group into its individual topics. The gap in male-stereotyped topics comes almost entirely from immigration: male candidates reduce its salience by 4.5 percentage points ($p < 0.10$), while female candidates leave

it unchanged (+0.9 percentage points), a contrast that matches the 5.4 percentage point gap reported for immigration in Table 2. Within female-stereotyped topics, women reduce the salience of health by 2.2 percentage points ($p < 0.05$) and, to a lesser extent, of education (−1.5 percentage points, not significant), whereas men leave health unchanged (+0.3) and reduce education by 1.8 percentage points ($p < 0.10$).

Taken together, these results suggest that the second-round leads candidates to de-emphasize the topics stereotypically associated with their own gender: men move away from immigration toward non-gendered topics, while women reduce their emphasis on health topics but maintain their emphasis on male-stereotyped ones. Note that the discontinuity design causally identifies only the differences between women and men; the side-specific averages are conditional means at the threshold and should be interpreted as descriptive evidence on the mechanism.

Table 4: Between-round change in topic salience at the cutoff, by candidate gender

Panel A: Aggregated topic groups

	(1) Female-stereotyped	(2) Male-stereotyped	(3) Other
Men: average change	-0.010 (0.011)	-0.039** (0.019)	0.044** (0.017)
Women: average change	-0.032*** (0.011)	0.012 (0.013)	0.001 (0.016)
Difference (women – men)	-0.022 (0.016)	0.050** (0.023)	-0.043* (0.023)
Bandwidth	3.331	2.349	2.732

Panel B: Individual topics

	(1) Health	(2) Education	(3) Justice	(4) Immigration	(5) Local	(6) National	(7) Other policy
Men: average change	0.003 (0.007)	-0.018* (0.010)	-0.011 (0.009)	-0.045* (0.026)	-0.005 (0.015)	0.111*** (0.040)	-0.054 (0.037)
Women: average change	-0.022** (0.009)	-0.015 (0.011)	0.006 (0.010)	0.009 (0.018)	0.005 (0.010)	0.055** (0.025)	-0.020 (0.032)
Difference (women – men)	-0.024** (0.012)	0.002 (0.014)	0.016 (0.013)	0.054* (0.032)	0.010 (0.018)	-0.056 (0.047)	0.034 (0.049)
Bandwidth	2.471	2.423	3.288	2.423	3.919	3.221	2.955

Notes: The outcome is the change in the salience of each topic group between the first- and second-round platforms. Panel A aggregates the topics into female-stereotyped (health and education), male-stereotyped (justice and criminality, and immigration), and remaining topics; Panel B reports the individual topics that make up the two stereotyped groups. Each column uses the bandwidth reported in Table 2. The rows Men and Women report the average change at the cutoff for male and female candidates who barely qualified, from a local linear regression with a triangular kernel and separate slopes on each side of the cutoff. Standard errors clustered by district-year in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Do candidates strategically adapt to the gender of their opponent?

Role congruity theory (Eagly & Karau, 2002) predicts that prejudice against female leaders arises from perceived incongruity between the female gender role and the leader role, and it activates when gender becomes salient. A mixed-gender runoff makes gender salient in a way that a same-gender runoff does not, so the constraints on how women may campaign should bind most when the opponent is a man. A related prediction comes from work on self-stereotyping: Coffman (2014) shows that individuals are less willing to contribute their ideas in domains that are incongruent with their gender, and that this reflects beliefs about their own ability rather than differences in actual ability, while Bordalo et al. (2019) show that such beliefs respond to the gender composition of the comparison group.

I examine whether candidates adjust their campaigns to their opponents' gender by running a heterogeneity analysis. Table 5 re-estimates the analysis separately for races in which the second-round opponent is a man and races in which the opponent is a woman. For male-stereotyped topics, the gap is 0.046 against a male opponent and 0.054 against a female opponent; therefore, the adjustment made by males is clearly not driven by the opponent's gender. For female-stereotyped topics and for the remaining topics, the subgroup estimates differ more. Women decrease the prevalence of female-stereotyped topics when they compete against a woman by 4.9 percentage points ($p = 0.057$), while when they compete against a man the change is approximately 0. Since splitting the sample roughly halves the effective number of observations in each cell, these tests have limited power; therefore, I test the results' robustness to different bandwidths and conclude that the effects are stable (Appendix Figure E.8).

In Table 6, I test whether female and male candidates change their rhetorical register. I do not find that, between the first and second round, candidates change their language to a more agentic or communal tone. During the first round, there were also no significant differences. However, while female candidates do not change their language when they compete against a woman (-0.019), they increase their use of communal language by 0.907 standard deviations when they compete against a man. In Appendix Figure E.9, I confirm the robustness of these results to different bandwidths.

Overall, the reduction in female-stereotyped topics only when candidates compete against a woman, and the increase in communal language when candidates compete against a man, point in favor of role congruity theory.

Table 5: Differences in topic adjustment between female and male candidate depending on the gender of the second-round opponent

Opponent	Female-stereotyped		Male-stereotyped		Other	
	Male	Female	Male	Female	Male	Female
Woman	-0.001 (0.037)	-0.049 (0.030)	0.046 (0.057)	0.054 (0.044)	-0.075 (0.059)	-0.004 (0.037)
Robust p -value	0.899	0.057	0.197	0.227	0.051	0.523
Observations	568	395	568	395	568	395
Eff. number of obs.	232	169	183	168	189	186
Polyn. order	1	1	1	1	1	1
Bandwidth	3.302	3.337	2.639	3.237	2.730	3.703
Mean, left of threshold	-0.031	0.017	-0.030	-0.046	0.062	0.018

Notes: The dependent variables take values between 0 and 1, and their computation is explained in Section 3.3.1. In Panel A, the variable of interest is being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. The heterogeneity analysis is performed following Calonico, Cattaneo, Farrell, Palomba, and Titiunik (2025). The mean gives the average outcome value for male candidates at the threshold.
^{*} $p < 0.10$, ^{**} $p < 0.05$, ^{***} $p < 0.01$.

Table 6: Agentic versus communal language

	(1)	(2)	(3)	
	1st round	Adjustment 1st to 2nd round	Adjustment, by opponent Male	Female
Woman	0.430 (0.267)	-0.481 (0.269)	-0.906 (0.632)	-0.020 (0.473)
Robust p -value	0.163	0.158	0.101	0.659
Observations	963	963	568	395
Eff. number of obs	355	297	188	138
Polyn. order	1	1	1	1
Bandwidth	2.969	2.381	2.683	2.695
Mean, left of threshold	-0.257	0.179	0.411	-0.119

Notes: The dependent variable measures the balance between agentic and communal language: higher values indicate more agentic language, lower values more communal language; the construction is explained in Subsection 3.3.3. Column (1) uses the first-round level, column (2) the change between the first and second round, and columns (3) the change between rounds estimated separately for races in which the second-round opponent is a man or a woman. Each column reports the results from a separate local polynomial regression. The variable of interest is being a female candidate rather than a male candidate. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. Columns (1) and (2) are estimated with `rdrobust` and columns (3) with the heterogeneity estimator of Calonico et al. (2025), which selects a bandwidth within each subgroup. Standard errors are clustered at the district-year level and shown in parentheses, and p -values are robust bias-corrected following Calonico et al. (2014). The mean gives the average outcome value for male candidates at the threshold.

6.3 Gender differences and electoral competitiveness

Experimental research shows that men and women respond differently to performance feedback in competitive environments (Berlin and Dargnies (2016)). Providing relative performance feedback allows to reduce gender differences in competitive choices (Wozniak, Harbaugh, and Mayr (2014), Baier, Davis, and Jaber-Lopez (2024)). Given the quasi-experimental nature of French legislative elections, first-round electoral results can serve as relative performance feedback for candidates.

To test whether a candidate's position in the race matters for adjustment, Table 7 exploits heterogeneity in victory margins. I define victory margin as the difference in the vote shares in the first-round between the candidate and the opponent of the second-round. In Panel A, I treat victory margin as a discrete variable and split the analysis into candidates whose victory margin is below the median and those whose victory margin is above the median. I conclude that the results are driven by the candidates whose victory margin is below the median, that is, whose candidates are clearly behind the opponent. When candidates are at a disadvantage relative to their opponents, the prevalence of male-stereotyped topics is 7.6 percentage points higher among female candidates than among males. When the victory margin is above the median, female candidates do not differ from male candidates on any topic. In Panel B, I use victory margin as a continuous variable, which summarizes the heterogeneity analysis. When women obtained the same share of votes as their opponents in the first-round, there are no significant differences. Each additional point of first-round disadvantage is associated with a 0.012 percentage-point increase in male-stereotyped topics. In the Appendix, Table E22 shows the same pattern for health: female candidates decrease the prevalence of health when they are at a disadvantage relative to their opponent.

6.4 Electoral outcomes

The results establish that between the first and second-round, female and male candidates reduce the prevalence of female- and male-stereotyped topics, respectively. A natural next question is whether this difference matters electorally: does adjusting the salience of gender-stereotyped topics affect the votes a candidate receives in the second-round?

Answering this question requires care. The discontinuity design identifies the effect of a female candidate, rather than a male candidate qualifying for the runoff, on both the candidate's platform and her vote share, but it does not identify the effect of the platform on vote share. To address this challenge, I follow Acharya et al. (2016) and estimate the average controlled direct effect (ACDE): the causal

Table 7: Differences in topics adjustment between female and male candidates - Heterogeneity by victory margin in the 1st round between the candidate and the opponent

	Female-stereotyped		Male-stereotyped		Other	
	Below	Above	Below	Above	Below	Above
Panel A: Heterogeneity with discretized victory margin						
Woman	-0.024 (0.029)	-0.020 (0.040)	0.076 (0.065)	0.034 (0.031)	-0.116 (0.074)	-0.028 (0.042)
P-value (below)=(above)	0.763		0.103		0.057	
Observations	481	482	481	482	481	482
Eff. number of obs	201	167	156	191	122	187
Robust p-value	0.242	0.624	0.056	0.831	0.025	0.940
Polyn. order	1	1	1	1	1	1
Bandwidth	3.095	2.951	2.419	3.288	1.766	3.227
Mean, left of threshold	-0.022	0.004	-0.063	-0.026	0.096	0.038
Panel B: Heterogeneity with linear victory margin						
Woman at mean margin	-0.016 (0.036)		-0.027 (0.045)		0.003 (0.046)	
Robust p-value	0.499		0.123		0.726	
Woman # victory margin	0.001 (0.003)		-0.012 (0.007)		0.007 (0.005)	
Robust p-value	0.841		0.003		0.014	
Observations	963		963		963	
Eff. number of obs	810		590		662	
Polyn. order	1		1		1	
Bandwidth	3.338		2.357		2.751	

Victory margin is the difference between the share of votes the candidate received in the first round and the opponent's share in the second-round. In Panel A, Below refers to candidates whose victory margin is below the median, while Above refers to candidates whose victory margin is above the median. In Panel B, victory margin is a continuous variable. Standard errors in parentheses are clustered at the district-year level. Each column reports the results from a separate local polynomial regression. The variable of interest is being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. The heterogeneity analysis is performed following Calonico et al. (2025). The dependent variables take values between 0 and 1, and their computation is explained in Section 3.3.1. The mean gives the average outcome value for male candidates at the threshold.

effect of a treatment (gender) after accounting for the effect of the mediator (topic adjustment between rounds).

The ACDE is estimated by sequential g-estimation. First, the outcome (the share of votes in the second-round) is regressed on the treatment, the measure of topic adjustment, pre-treatment covariates, and any intermediate confounders;

the estimated effect of the adjustment is then subtracted from the outcome.¹³ In a second stage, the demediated outcome is regressed on the treatment and pre-treatment covariates alone. Standard errors are obtained by a block bootstrap at the district-year level, which accounts for the fact that the demediated outcome depends on a first-stage estimate.¹⁴

Two assumptions are required. The first, that candidate gender is as good as randomly assigned near the threshold, is delivered by the discontinuity design. The second is that the between-round adjustment is as good as random, conditional on candidate gender, pre-treatment covariates, and intermediate confounders.

Table 8 reports the results. Panel A shows no gender gap in votes, and between-round adjustments account for none of the difference in electoral performance between male and female candidates. The total effect of a woman rather than a man qualifying for the runoff on second-round vote share is small and statistically insignificant (0.003 percentage points, standard error 0.007), and fixing both topic adjustments at zero for all candidates leaves it essentially unchanged (0.001, standard error 0.007). The difference between the two, 0.002 points, is not statistically distinguishable from zero.

Panel B shows why: the adjustments themselves have no detectable effect on vote share. A one-percentage-point increase in the salience of male-stereotyped topics changes a female candidate's second-round vote share by 0.013 points (not significant), and the corresponding effect for female-stereotyped topics is also small. The effects do not differ appreciably between male and female candidates, so there is no evidence that voters reward or penalize the same adjustment differently by gender.

The absence of electoral returns is informative about the behavior documented in this paper. If candidates adjusted their platforms to maximize votes, they would concentrate adjustment where it pays off. Instead, male and female candidates adjust along gender lines, even though the adjustment has no detectable effect on their electoral performance. This pattern is more consistent with candidates acting on shared expectations about how men and women are supposed to campaign than with a vote-maximizing strategy that exploits differences in how voters evaluate male and female candidates.

¹³Intermediate confounders are variables that are affected by the treatment and affect both the mediator and outcome. I do not include any intermediate confounders, since all the potential controls are pre-treatment.

¹⁴All estimates use the MSE-optimal bandwidth for second-round vote share (Calonico et al., 2014), held fixed across bootstrap replications.

Table 8: Electoral consequences of between-round platform adjustment

	(1)	(2)
	Female-stereotyped	Male-stereotyped
<i>Panel A: Gender gap in 2nd round vote share</i>		
Total effect	0.003 (0.007)	
Controlled direct effect	0.001 (0.007)	
Difference	0.002 (0.002)	
<i>Panel B: Effect of adjustment on 2nd round vote share</i>		
Female candidates	-0.055 (0.047)	0.013 (0.028)
Male candidates	-0.024 (0.041)	0.006 (0.024)
Observations	372	
Bandwidth	3.077	

Notes: Estimates from sequential g-estimation (Acharya et al., 2016), using local linear regressions with a triangular kernel and the MSE-optimal bandwidth for second-round vote share. The outcome is the second-round vote share and the mediators are the between-round changes in the salience of female- and male-stereotyped topics, all measured as shares between 0 and 1. Columns (1) and (2) come from a single model in which the two topic mediators enter jointly, with non-gendered topics as the omitted category. The first stage regresses the outcome on the treatment, the mediators, their interactions with the treatment, and pre-treatment covariates. The second stage regresses the demediated outcome on the treatment and pre-treatment covariates only. Panel A compares the total effect with the controlled direct effect, which fixes the adjustments at zero for all candidates. Panel B reports the effect on vote share of a one-unit change in each adjustment. In Panels A and B, standard errors in parentheses are from 1,000 cluster-bootstrap replications at the district-year level, and significance stars use a normal approximation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.5 Discussion of alternative mechanisms

The evidence presented above is consistent with men adjusting to broaden the topics discussed in the second-round, while women adjust to conform to stereotypes. I now discuss alternative interpretations for the findings.

Differences in policy preferences Gender differences in campaigns could simply reflect gender differences in policy preferences, meaning that female candidates might decide to decrease the salience of health topics because they give less value to them. However, this rationality is against the findings of the literature on elected women. Lippmann (2022) finds that female elected politicians are more active on women’s issues while men on military issues. Similarly, Baskaran and Hessami (2025) conclude that female councilors promote the expansion of public child care. Hessami and da Fonseca (2020) provide a literature review on female political representation and effects on policies, and report no significant differences in public spending. Even if in the specific context of French legislative elections,

female candidates have a preference for policies related to justice & criminality and immigration, why did they do not reveal their preferences in the first-round? Table 1 shows no significant differences, with the effects very close to zero.

Differences in campaign donations If campaign funds are essential to a politician's success, then large differences in how much money male and female candidates raise might affect how women and men campaign in their races. Previous research has causally shown a gender gap in campaign fundraising for US state legislators (Barber et al. (2016)) and Brazilian mayoral candidates (Brollo and Troiano (2016), Ferraz, Nogueira, and Tavares (2022)). I exploit differences in contributions from the party, donors, or personal resources to disentangle party and donor discrimination from differences in personal resources. In Table E23, I show no significant differences between female and male candidates in any campaign resource. These results have several implications. First, the lack of differences in total expenditures or revenues implies that women are not more or less financially constrained. Second, personal contributions are also a proxy for financial risk-aversion, and the lack of significant differences suggests that gender differences in risk-aversion are unlikely to explain the results. Third, no significant differences in party contributions show that parties do not treat female and male candidates differently. Fourth, donors do not seem to discriminate between female and male candidates, although the sources might differ and such information is unavailable.

Candidates adjust to the gender of absent candidates The RDD compares female candidates who barely qualified for the second round (while a male candidate narrowly failed to qualify in that electoral district) with male candidates who barely qualified (while a female candidate narrowly failed to qualify). The treatment therefore bundles the finalist's gender with the gender of the candidate who was narrowly eliminated. If the eliminated candidate's identity matters, both female and male finalists may respond to having narrowly beaten a candidate of the opposite gender. For example, if female candidates ran in the first round but only male candidates remained in the second, male finalists might try to attract the eliminated female candidate's voters by downplaying male-stereotyped topics.

I examine whether the effects vary with the electoral weight of the eliminated female candidates. I measure it as the sum of the first-round vote shares, expressed as a percentage of registered voters, of all female candidates in the electoral district who did not advance to the second round. Because the number of female candidates, and hence their combined vote share, increased substantially over the period, particularly after the 2000 parity law, I demean this variable by election year. Thus, it captures whether eliminated female candidates were electorally stronger than in other districts in the same election, rather than differences across election

years. If finalists court the voters of eliminated women, the adjustment should be stronger where these voters are more numerous.

Table E24 reports the results. The Woman row gives the effect at the election-year mean of the moderator, and the interaction term gives how the effect changes with each additional percentage point of the vote share of eliminated women. None of the interaction terms is statistically significant. I therefore find no evidence that finalists tailor their programs to the voters of eliminated female candidates.

7 Conclusion

Using individual campaign platforms covering all French legislative elections from 1981 to 2024, I provide the first causal evidence of gender differences in campaign strategies. The two-round electoral system enables a regression discontinuity design that compares female candidates who barely qualify for the second round with male candidates in nearly identical electoral environments. While during the first-round of the campaign, female and male candidates do not differ on the issues discussed or how they discuss these issues, gender differences emerge during the second-round of the political campaign. I find that between the first and second-round, there is a higher prevalence of male-stereotyped topics and a lower prevalence of female-stereotyped topics among women, in comparison to males. These adjustments stem from each gender decreasing the prevalence of its respective stereotyped topic: male candidates decrease the prevalence of male-stereotyped topics, while women decrease the prevalence of female-stereotyped topics.

I provide suggestive evidence that female candidates adjust their behavior to the opponent. They reduce the prevalence of female-stereotyped topics only when they compete against a woman; by contrast, males' reduction of male-stereotyped topics seems driven more by increasing the prevalence of non-policy issues. I also try to understand how candidates adjust their behavior in terms of agentic vs communal language. Agentic language allows candidates to present themselves as leaders and results-driven, while communal language positions candidates as individuals willing to listen and close to their constituents. According to stereotypes, women are expected to use more communal than agentic language. Although female candidates do not use a different language from males when they compete against a woman, they use more communal language when they compete against a man.

These results have implications for how we think about political representation. Most of what we know about gender and politics concerns elected officials: what they legislate, how they allocate budgets, and whether their presence en-

courages other women to run. The evidence here points to a margin that operates earlier. Because the issues a campaign makes salient help determine the terms of an election, the gender composition of a race shapes political debate even when no woman is ultimately elected. A parity law or a candidate selection rule that changes who competes therefore changes campaign content, independently of whether it changes who wins.

The paper opens several avenues for research. First, we need to better understand the beliefs that drive politicians' behavior. This paper infers them from choices and the absence of electoral returns; we need further evidence on what candidates believe voters reward, and what voters reward by gender. Such evidence would let us distinguish anticipating a penalty from norms irrespective of the penalty. Second, while this paper concludes that in the first-round female and male candidates do not significantly differ in how they campaign, and that the differences that emerge in the second-round are driven either by strategic behavior or norms, Lippmann (2022) finds that elected female politicians are most active on women's issues while men are involved in military issues. We need to better understand whether a politician's campaign behavior lets voters predict their behavior once elected, or whether elected politicians' behavior is also driven by peers and party constraints.

References

- Abele, A. E., & Wojciszke, B. (2007). Agency and communion from the perspective of self versus others. *Journal of personality and social psychology*, 93(5), 751.
- Acharya, A., Blackwell, M., & Sen, M. (2016). Explaining causal findings without bias: Detecting and assessing direct effects. *American Political Science Review*, 110(3), 512–529.
- Alesina, A. (1988). Credibility and policy convergence in a two-party system with rational voters. *The American Economic Review*, 78(4), 796–805.
- Antoun, W., Kulumba, F., Touchent, R., de la Clergerie, É., Sagot, B., & Seddah, D. (2024). Camembert 2.0: A smarter french language model aged to perfection. *arXiv preprint arXiv:2411.08868*.
- Bagues, M., & Campa, P. (2021). Can gender quotas in candidate lists empower women? evidence from a regression discontinuity design. *Journal of Public Economics*, 194, 104315.
- Baier, A., Davis, B., & Jaber-Lopez, T. (2024). Gender, choice of task, and the effect of feedback on competition: An experiment. *Journal of Economic Psychology*, 103, 102743.
- Bakker, B. N., Schumacher, G., & Rooduijn, M. (2021). Hot politics? affective responses to political rhetoric. *American Political Science Review*, 115(1), 150–164.
- Barber, M., Butler, D. M., Preece, J., et al. (2016). Gender inequalities in campaign finance. *Quarterly Journal of Political Science*, 11(2), 219–48.
- Baskaran, T., & Hessami, Z. (2018). Does the election of a female leader clear the way for more women in politics? *American Economic Journal: Economic Policy*, 10(3), 95–121.
- Baskaran, T., & Hessami, Z. (2025). Women in political bodies as policymakers. *Review of Economics and Statistics*, 107(6), 1501–1517.
- Beaman, L., Chattopadhyay, R., Duflo, E., Pande, R., & Topalova, P. (2009). Powerful women: does exposure reduce bias? *The Quarterly journal of economics*, 124(4), 1497–1540.
- Bekkouche, Y., Cagé, J., & Dewitte, E. (2022). The heterogeneous price of a vote: Evidence from multiparty systems, 1993–2017. *Journal of Public Economics*, 206, 104559.
- Berlin, N., & Dargnies, M.-P. (2016). Gender differences in reactions to feedback and willingness to compete. *Journal of Economic Behavior & Organization*, 130, 320–336.
- Besley, T., & Coate, S. (1997). An economic model of representative democracy.

- The Quarterly Journal of Economics*, 112(1), 85–114.
- Bhalotra, S., & Clots-Figueras, I. (2014). Health and the political agency of women. *American Economic Journal: Economic Policy*, 6(2), 164–197.
- Bhavnani, R. R. (2009). Do electoral quotas work after they are withdrawn? evidence from a natural experiment in india. *American Political Science Review*, 103(1), 23–35.
- Bordalo, P., Coffman, K., Gennaioli, N., & Shleifer, A. (2019). Beliefs about gender. *American Economic Review*, 109(3), 739–773.
- Brollo, F., & Troiano, U. (2016). What happens when a woman wins an election? evidence from close races in brazil. *Journal of Development Economics*, 122, 28–45.
- Calonico, S., Cattaneo, M. D., Farrell, M. H., Palomba, F., & Titiunik, R. (2025). Treatment effect heterogeneity in regression discontinuity designs. *arXiv preprint arXiv:2503.13696*.
- Calonico, S., Cattaneo, M. D., Farrell, M. H., & Titiunik, R. (2019). Regression discontinuity designs using covariates. *Review of Economics and Statistics*, 101(3), 442–451.
- Calonico, S., Cattaneo, M. D., & Titiunik, R. (2014). Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica*, 82(6), 2295–2326.
- Casarico, A., Lattanzio, S., & Profeta, P. (2022). Women and local public finance. *European Journal of Political Economy*, 72, 102096.
- Cattaneo, M. D., Jansson, M., & Ma, X. (2018). Manipulation testing based on density discontinuity. *The Stata Journal*, 18(1), 234–261.
- Chattopadhyay, R., & Duflo, E. (2004). Women as policy makers: Evidence from a randomized policy experiment in india. *Econometrica*, 72(5), 1409–1443.
- Coffman, K. B. (2014). Evidence on self-stereotyping and the contribution of ideas. *The Quarterly Journal of Economics*, 129(4), 1625–1660.
- Di Tella, R., Kotti, R., Le Pennec, C., & Pons, V. (2025). Keep your enemies closer: strategic platform adjustments during us and french elections. *American Economic Review*, 115(8), 2488–2528.
- Downs, A. (1957). An economic theory of political action in a democracy. *Journal of Political Economy*, 65(2), 135–150.
- Eagly, A. H., & Karau, S. J. (2002). Role congruity theory of prejudice toward female leaders. *Psychological Review*, 109(3), 573.
- Esteve-Volart, B., & Bagues, M. (2012). Are women pawns in the political game? evidence from elections to the spanish senate. *Journal of public Economics*, 96(3-4), 387–399.

- Ferraz, C., Nogueira, C., & Tavares, J. (2022). Does gender-targeted electoral funding work? evidence from brazil.
- Ferreira, F., & Gyourko, J. (2014). Does gender matter for political leadership? the case of us mayors. *Journal of Public Economics*, 112, 24–39.
- France. presidential election 10 and 24 april 2022. odhr election assessment mission final report. (2022).
- Funk, P., & Gathmann, C. (2015). Gender gaps in policy making: Evidence from direct democracy in switzerland. *Economic Policy*, 30(81), 141–181.
- Gaultier-Voituriez, O. (2016). Archelec, les archives électorales de la v e république, du papier au numérique. *Histoire@ Politique*, 3, 213–220.
- Gennaro, G., & Ash, E. (2022). Emotion and reason in political language. *The Economic Journal*, 132(643), 1037–1059.
- Gneezy, U., Niederle, M., & Rustichini, A. (2003). Performance in competitive environments: Gender differences. *The Quarterly Journal of Economics*, 118(3), 1049–1074.
- Gneezy, U., & Rustichini, A. (2004). Gender and competition at a young age. *American Economic Review*, 94(2), 377–381.
- Granzier, R., Pons, V., & Tricaud, C. (2023). Coordination and bandwagon effects: How past rankings shape the behavior of voters and candidates. *American Economic Journal: Applied Economics*, 15(4), 177–217.
- Hessami, Z., & da Fonseca, M. L. (2020). Female political representation and substantive effects on policies: A literature review. *European Journal of Political Economy*, 63, 101896.
- Huddy, L., & Terkildsen, N. (1993). Gender stereotypes and the perception of male and female candidates. *American journal of political science*, 119–147.
- Iriberri, N., & Rey-Biel, P. (2017). Stereotypes are only a threat when beliefs are reinforced: On the sensitivity of gender differences in performance under competition to information provision. *Journal of Economic Behavior & Organization*, 135, 99–111.
- Jolly, S., Bakker, R., Hooghe, L., Marks, G., Polk, J., Rovny, J., ... Vachudova, M. A. (2022). Chapel hill expert survey trend file, 1999–2019. *Electoral studies*, 75, 102420.
- Kanthak, K., & Woon, J. (2015). Women don't run? election aversion and candidate entry. *American journal of political science*, 59(3), 595–612.
- Le Barbanchon, T., & Sauvagnat, J. (2022). Electoral competition, voter bias, and women in politics. *Journal of the European Economic Association*, 20(1), 352–394.
- Le Pennec, C. (2023). Strategic campaign communication: Evidence from 30,000

candidate manifestos. *The Economic Journal*, uead082.

Lippmann, Q. (2021). Are gender quotas on candidates bound to be ineffective? *Journal of Economic Behavior & Organization*, 191, 661–678.

Lippmann, Q. (2022). Gender and lawmaking in times of quotas. *Journal of Public Economics*, 207, 104610.

Marshall, J. (2024). Can close election regression discontinuity designs identify effects of winning politician characteristics? *American Journal of Political Science*, 68(2), 494–510.

Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems*, 26.

Niederle, M., & Vesterlund, L. (2007). Do women shy away from competition? do men compete too much? *The Quarterly Journal of Economics*, 122(3), 1067–1101.

Osborne, M. J., & Slivinski, A. (1996). A model of political competition with citizen-candidates. *The Quarterly Journal of Economics*, 111(1), 65–96.

Osnabrügge, M., Hobolt, S. B., & Rodon, T. (2021). Playing to the gallery: Emotive rhetoric in parliaments. *American Political Science Review*, 115(3), 885–899.

Piolat, A., Booth, R. J., Chung, C. K., Davids, M., & Pennebaker, J. W. (2011). La version française du dictionnaire pour le liwc: modalités de construction et exemples d'utilisation. *Psychologie Française*, 56(3), 145–159.

Rudman, L. A., & Glick, P. (2001). Prescriptive gender stereotypes and backlash toward agentic women. *Journal of social issues*, 57(4), 743–762.

Wozniak, D., Harbaugh, W. T., & Mayr, U. (2014). The menstrual cycle and performance feedback alter gender differences in competitive choices. *Journal of Labor Economics*, 32(1), 161–198.

Appendix

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A Institutional context

I assign candidates to seven political orientations (far-left, left, liberal, right, far-right, and other). I use the party classifications from the Chapel Hill Expert Survey (CHES) and, when missing, from the ParlGov and Granzier et al. (2023). I measure party positioning on a scale from 0 (left) to 1 (right) and include family classification.

1981 Parliamentary Elections	
Political label	Political orientation
Parti Communiste Français	Far-left
Divers Droite	Right
Divers Gauche	Left
Ecologistes	Left
Extrême Droite	Far-right
Extrême Gauche	Far-left
Indépendants	Other
Non Classés	Other
Rassemblement pour la République	Right
Socialistes	Left
Union pour la Démocratie Française	Right
1988 Parliamentary Elections	
Political label	Political orientation
Parti Communiste Français	Far-left
Divers Droite	Right
Ecologistes	Left
Extrême Droite	Far-right
Extrême Gauche	Far-left
Front National	Far-right
Majorité Présidentielle	Left
Radicaux de Gauche	Left
Régionalistes	Other
Rassemblement pour la République	Right
Socialistes	Left
Union pour la Démocratie Française	Right

1993 Parliamentary Elections	
Political label	Political orientation
Parti Communiste Français	Far-left
Divers	Other
Divers Droite	Right
Extrême Droite	Far-right
Extrême Gauche	Far-left
Front National	Far-right
Génération Ecologie	Left
Majorité Présidentielle	Left
Radicaux de Gauche	Left
Régionalistes	Other
Rassemblement pour la République	Right
Parti Socialiste	Left
Union pour la Démocratie Française	Right
Les Verts	Left

1997 Parliamentary Elections	
Political label	Political orientation
Parti Communiste Français	Far-left
Divers	Other
Divers Droite	Right
Divers Gauche	Left
Ecologistes	Left
Extrême Droite	Far-right
Extrême Gauche	Far-left
Front National	Far-right
Mouvement Pour la France	Far-right
Parti Radical Socialiste	Left
Rassemblement pour la République	Right
Socialistes	Left
Union pour la Démocratie Française	Right

2002 Parliamentary Elections	
Political label	Political orientation
Communistes	Far-left
Chasse, Pêche, Nature et Traditions	Right
Divers	Other
Démocratie Libérale	Right
Divers Droite	Right
Divers Gauche	Left
Ecologistes	Left
Extrême Droite	Far-right
Extrême Gauche	Far-left
Front National	Far-right
Ligue Communiste Révolutionnaire	Far-left
Lutte Ouvrière	Far-left
Mouvement des Citoyens	Left
Mouvement National Républicain	Far-right
Mouvement pour la France	Right
Pôle Républicain	Left
Radicaux de Gauche	Left
Régionalistes	Other
Rassemblement pour la France	Right
Socialistes	Left
Union pour la Démocratie Française	Liberal
Union pour un Mouvement Populaire	Right
Les Verts	Left

2007 Parliamentary Elections	
Political label	Political orientation
Communistes	Far-left
Chasse, Pêche, Nature et Traditions	Right
Divers	Other
Divers Droite	Right
Divers Gauche	Left
Ecologistes	Left
Extrême Droite	Far-right
Extrême Gauche	Far-left
Front National	Far-right
Majorité Présidentielle	Right
Mouvement pour la France	Right
Radicaux de Gauche	Left
Régionalistes	Other
Rassemblement pour la France	Right
Socialistes	Left
Union pour la Démocratie Française - Mouvement Démocrate	Liberal
Union pour un Mouvement Populaire	Right
Les Verts	Left

2012 Parliamentary Elections	
Political label	Political orientation
Alliance Centriste	Liberal
Autres	Other
Centre pour la France	Liberal
Communistes	Far-left
Divers Droite	Right
Divers Gauche	Left
Ecologistes	Left
Extrême Droite	Far-right
Extrême Gauche	Far-left
Front de Gauche	Far-left
Front National	Far-right
Nouveau Centre	Liberal
Parti Radical	Right
Radicaux de Gauche	Left
Régionalistes	Other
Socialistes	Left
Union pour un Mouvement Populaire	Right
Europe Ecologie - Les Verts	Left

2017 Parliamentary Elections	
Political label	Political orientation
Parti Communiste Français	Far-left
Debout la France	Far-right
Divers	Other
Divers Droite	Right
Divers Gauche	Left
Europe Écologie les Verts	Left
Extrême Droite	Far-right
Extrême Gauche	Far-left
France Insoumise	Far-left
Front National	Far-right
Les Républicains	Right
Modem	Liberal
Radicaux de Gauche	Left
Régionalistes	Other
République en Marche	Liberal
Parti Socialiste	Left
Union des Démocrates et Indépendants	Liberal
2022 Parliamentary Elections	
Political label	Political orientation
Divers	Other
Divers Centre	Other
Divers Droite	Right
Divers Gauche	Left
Divers Extrême-droite	Far-right
Divers Extrême-gauche	Far-left
Droite Souveraniste	Far-right
Écologiste	Left
Ensemble	Liberal
Extrême-droite	Far-right
Les Républicains	Right
Nupes	Left
Radicaux de Gauche	Left
Réconquête	Far-right
Régionalistes	Other
Rassemblement National	Far-right
Union des Démocrates et Indépendants	Liberal

2024 Parliamentary Elections	
Political label	Political orientation
Divers	Other
Divers Centre	Other
Divers Droite	Right
Divers Gauche	Left
Divers Extrême-droite	Far-right
Divers Extrême-gauche	Far-left
Droite Souveraniste	Far-right
Écologiste	Left
Ensemble	Liberal
Extrême-droite	Far-right
Horizons	Right
Les Républicains	Right
Nupes	Left
Radicaux de Gauche	Left
Réconquête	Far-right
Régionalistes	Other
Rassemblement National	Far-right
Union des Démocrates et Indépendants	Liberal
Union de Gauche	Left
Union Extrême-droite	Far-right

B Data

B.1 Political platforms

I transform the pdf versions of the manifestos into text using optical character recognition: Mistral OCR.

I perform the following pre-processing procedures.

I remove words containing party names, party acronyms, parliamentary titles, and terms describing blocs of parties. I also eliminate first and last names.

I pre-process the corpus following standard natural language processing practices: remove punctuation and numbers, convert all letters to lowercase, lemmatize each word, and restrict the vocabulary to words used in at least 1% and 50% of the documents, using the Spacy French version 3.5.0. Because OCR sometimes fails to detect accents, I convert words from Latin-1 to UTF-8 to reduce errors.

Some politicians opt for using the party platform instead of a personalized plat-

form; this is common in small parties (e.g., Rassemblement National and Green parties) but an infrequent practice among well-established parties (e.g., socialist and republican parties). I do not include manifestos similar to the party platform in the training datasets. Keeping duplicate measures introduces multicollinearity, making these manifestos count more. I include them in the regression model.

Table B1: Sampling frame

Year	First round			Second round		
	Races	Total candidates	Platforms collected	Races	Total candidates	Platforms collected
1981	491	2650	2459	334	660	652
1988	577	2849	2651	455	899	852
1993	577	5290	4119	497	992	964
1997	577	6359	2908	565	1197	1097
2002	577	8444	201	519	1045	55
2007	577	7634	395	467	933	80
2012	577	6603	862	541	1101	518
2017	577	7877	5768	573	1146	927
2022	577	6289	4902	561	1126	1059
2024	577	4009	2754	501	1094	850

Notes: The table shows the number of races, total candidates, and manifestos collected for each legislative election in the data set.

Figure B.1: Florence Blatrix-Contat political platform during 1st round 2017 legislative elections - 1st page

Florence BLATRIX-CONTAT

Votre Députée

Michel FONTAINE
suppléant

A gauche pour faire réussir la France

51 ans, mariée, 3 enfants

Je vis dans mon village natal à Drom, au cœur du Revermont.

Mes grands-parents agriculteurs et mes parents m'ont transmis leur attachement à ces terres ; ils m'ont surtout appris que l'école était un moyen d'émancipation et d'ascension sociale.

Après des études comptables, je suis devenue enseignante. Agrégée d'économie-gestion, j'enseigne l'économie et le droit.

Elue depuis 1995 dans ma commune, puis en 2015 à la Région Auvergne-Rhône-Alpes, je suis convaincue que l'action publique et la volonté permettent d'avancer. Dans nos collectivités, je suis attachée à travailler avec tous, au-delà des clivages partisans ; c'est grâce au rassemblement des élus que les projets peuvent aboutir.

Madame, Monsieur,

Le 7 mai dernier, le peuple français a fait le choix d'un nouveau président de la République.

Les élections législatives des 11 et 18 juin décideront de l'avenir de la France ; elles seront l'occasion de choisir la République que nous voulons. Je me présente à vos suffrages avec la volonté de faire réussir la France, de faire réussir ce quinquennat. La majorité issue de ce scrutin devra agir pour la justice sociale, la solidarité, l'égalité des chances et la transition écologique. **Pour cela, la gauche doit être forte à l'Assemblée nationale.**

Demain, à l'Assemblée, je veux représenter une gauche constructive et exigeante. **Je veillerai à la défense des acquis sociaux, au respect du dialogue social et me battrai pour un modèle de croissance compatible avec la préservation de la planète.**

Avec mon suppléant, **Michel FONTAINE**, nous vivons, travaillons et sommes investis dans cette circonscription. Nous mettrons notre expérience à votre service pour développer sur ce territoire : l'emploi, les services publics, la santé, la solidarité, mais aussi les activités culturelles et sportives indispensables au lien social. Elue rurale, je sais à quel point les services publics doivent être préservés dans chaque commune.

Sur le terrain comme à l'Assemblée, je mettrai toute mon énergie pour accompagner et soutenir les projets de notre territoire et pour aider celles et ceux qui rencontrent des difficultés.

Dimanche 11 juin, je compte sur votre soutien. Vous pouvez compter sur ma détermination et mon dévouement.

Florence Blatrix Contat

Source: RegardsCitoyens (<https://github.com/regardscitoyens>)

Figure B.2: Florence Blatrix-Contat political platform during 1st round 2017 legislative elections - 2nd page



Michel FONTAINE

*Premier-adjoint de la Ville de Bourg-en-Bresse
Président de l'Agglomération de 2008 à fin 2016
et depuis Vice-président au développement économique.*

Je vis sur Bourg-en-Bresse depuis plus de 40 ans, j'ai partagé ma vie professionnelle entre le lycée Carriat et ma société FontainePicard.

J'ai deux enfants et vis maritalement. J'ai assumé des responsabilités associatives avant de devenir un élu local.

En m'engageant aux côtés de Florence Blatrix-Contat, je fais le choix de la réussite dans une grande fidélité à mes convictions.

NATIONALEMENT, je serai une députée constructive mais exigeante vis-à-vis des réformes proposées par le président de la République, fidèle aux valeurs qui fondent mon engagement politique à Gauche.

Je soutiendrai

- **La moralisation de la vie publique et l'interdiction du cumul** de plus de trois mandats successifs pour permettre le renouvellement
- Les projets visant à concilier **efficacité économique et justice sociale** ainsi qu'une action forte en matière de **formation pour lutter contre le chômage**
- La priorité donnée à l'école primaire pour que chaque enfant puisse réussir et le soutien à l'autonomie des jeunes par une **revalorisation des bourses**
- Les mesures en faveur du pouvoir d'achat pour les plus modestes avec **l'augmentation du minimum vieillesse et de la prime d'activité**
- **La transition écologique et énergétique** : des investissements dans la croissance verte et les économies d'énergie, l'interdiction des perturbateurs endocriniens

Je serai vigilante

- Sur la justice fiscale : **je m'opposerai à la baisse de l'Impôt Sur la Fortune et à l'augmentation de la CSG** qui entraînera une diminution des pensions pour plus de 8 millions de retraités
- Sur le dialogue social et les droits des salariés : **le dialogue social doit être renforcé** et le Code du travail doit rester un socle protecteur élevé pour les salariés
- Sur les services publics : « Ils sont le patrimoine de ceux qui n'en ont pas » ; je m'opposerai à une nouvelle baisse du nombre de fonctionnaires et des **dotations des collectivités**

LOCALEMENT, je défendrai les projets de nos territoires : fibre optique, investissement dans l'université, dans les activités sportives et culturelles, lutte contre les déserts médicaux et maintien des services en milieu rural.

« Ce quinquennat sera réussi si la France n'oublie personne en route. Pour cela, nous avons besoin d'une gauche forte. Je sais que Florence et Michel portent ces valeurs. Je vous appelle à les soutenir les 11 et 18 juin prochains. »



Jean-François DEBAT

Candidate de la gauche, des démocrates et des écologistes, je compte sur vous dès le 1^{er} tour.



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(continuation)

B.2 Measurement

B.2.1 Topics: CamemBERT

Table B2: Prompt shared with ChatGPT to classify a sample of paragraphs into topics

You are an expert in French political science. You must classify political paragraphs into EXACTLY ONE OR TWO categories.

Rules:

- Indicate the most obvious category. If there are several, also indicate the second most obvious one. Respond with a comma-separated list.
- No comments
- No additional text
- Choose only from the following categories:

“Immigration”: immigration, migrants, borders, asylum, citizenship.

Example: “We need to strengthen border controls to limit illegal immigration.”

“Justice and Crime”: police, justice, crime, defense, terrorism, courts, prisons.

Example: “The number of burglaries has increased in suburban areas.”

“Health”: health care, hospitals, medicine, public health, health safety.

Example: “Medical deserts are affecting more and more rural counties.”

“Economy”: public finances, taxation, budget, deficit, fiscal policy, public debt, taxes, VAT, public spending, fiscal and monetary policy, businesses, industry, trade, markets, economic growth, exports, entrepreneurship, SMEs.

Example: “The budget deficit exceeds 3% of GDP this year.”

Example: “The reindustrialization of the region is a priority for our SMEs.”

“Environment”: climate, ecology, energy, pollution, agriculture, biodiversity, nuclear power, renewable energy, energy transition.

Example: “France must accelerate its transition to renewable energy.”

“Education”: schools, universities, teaching, training, students, scientific research.

Example: “The shortage of teachers in rural areas is a cause for concern.”

“Employment and Social Security”: work, unemployment, wages, pensions, labor relations, unions, social protection.

Example: “The pension reform plans to raise the legal retirement age to 64.”

“Foreign Policy”: foreign policy, Europe, international relations, diplomacy, international treaties.

Example: “France must strengthen its role within European institutions.”

"Local politics": sentences relating to local authorities, municipalities, departments, regions, mayors, municipal or regional elections, decentralization, and local powers.

Example: "Mayors are demanding greater administrative autonomy for their municipalities."

"Politics": sentences relating to the functioning of national political institutions, political parties, the results of legislative or presidential elections, relations between the government and the opposition, parliamentary procedures, political alliances, and election polls.

Example: "I ask you to remain mobilized until Sunday for these elections, the most important of the Fifth Republic."

"Other": Does not clearly fall under any of the above categories. In this case, simply indicate that this is the only category.

Table B3: Topics per-class performance

Topic	Precision	Recall	F1-score
Economy	0.86	0.89	0.88
Employment & social security	0.87	0.90	0.88
Environment	0.90	0.90	0.90
Health	0.93	0.83	0.88
Education	0.85	0.86	0.86
Justice & criminality	0.87	0.87	0.87
Immigration	0.85	0.80	0.82
Foreign policy	0.80	0.88	0.84
Local politics	0.87	0.88	0.88
National politics	0.89	0.89	0.89
Other	0.86	0.81	0.83

Notes: Per-class performance comparison between the training and test samples. Scores range from 0 to 1, where 1 indicates perfect agreement.

Table B4: Topics per-class performance: final topics classification vs human annotator

Topic	Precision	Recall	F1-score
Female-stereotyped	0.682	0.824	0.746
Male stereotyped	0.782	0.748	0.764
Other	0.953	0.961	0.957

Notes: Per-class performance comparison between the final topic classification and a human annotator. Scores range from 0 to 1, where 1 indicates perfect agreement. The comparison is made for a random sample of 1000 paragraphs.

Table B5: Topics per-class performance: final topics classification vs LLM classification

Topic	Precision	Recall	F1-score
Female-stereotyped	0.764	0.816	0.789
Male stereotyped	0.836	0.760	0.797
Other	0.988	0.948	0.967

Notes: Per-class performance comparison between the final topic classification and an LLM classifier (ChatGPT 5.4 version). Scores range from 0 to 1, where 1 indicates perfect agreement. The comparison uses a random sample of 1000 paragraphs, the same sample used to evaluate per-class performance between the final topic classification and the human annotator.

Table B6: Topics per-class performance: human annotator vs LLM classification

Topic	Precision	Recall	F1-score
Female-stereotyped	0.718	0.813	0.763
Male stereotyped	0.736	0.774	0.754
Other	0.940	0.988	0.963

Notes: Per-class performance comparison between the human annotator and an LLM classifier (ChatGPT 5.4 version). Scores range from 0 to 1, where 1 indicates perfect agreement. The comparison uses a random sample of 1000 paragraphs, the same sample used to evaluate per-class performance between the final topic classification and the human annotator.

B.2.2 Emotion vs reason

Table B7: Words that are closest to the poles of affect/emotion and rationality/cognition

Emotion	immorality, deprivation, marginality, rhyme, sadness, innocent, daddy, pornography, prey, uncertainty, horror, boredom, rosy, Nazism, Buchenwald, unconsciousness, despair, despairing, rage, install <i>immoralite, denuement, marginalite, rimer, tristesse, innocent, papu, pornographi, proie, incertitud, horreur, ennui, roser, nazisme, buchenwald, inconscience, desespoir, desesperant, rage, install</i>
Reason	judicious, will ensure, session, will render, sectoral, deliberation, lobbying, validate, article, publish, procurator, detailed, elaborate, in-depth, suggest, elaboration, previously, expressly, decisional, tripartite <i>judicieux, assurerai, session, rendrai, sectoriel, deliberation, lobbying, valider, articl, publier, procuratier, detaille, elaborer, approfondie, suggerer, elaboration, prealablement, expressement, decisionnel, tripartite</i>

B.2.3 Agentic vs communal language

Table B8: Prompt shared with ChatGPT to classify a sample of paragraphs into agentic language

You are an expert in French political science. You are given a segment (a sentence or a short paragraph) taken from the campaign platform of a candidate in the French legislative elections.

Question: In this segment, does the candidate present himself or herself in an AGENTIC register?

The agentic register emphasizes action, strength, and self-control: determination, tenacity, courage, firmness, the ability to make decisions, to lead, to achieve results, and to win; commitments expressed in an affirmative and confident manner ("I will do," "I will achieve," "I will not give up").

Important: Rate the WAY in which the candidate presents themselves, not the TOPIC addressed. The same theme can be addressed in different registers:

- "I will fight tirelessly to keep the maternity ward open" → 1
- "The maternity ward must remain open" → 0 (neutral programmatic statement)
- "I will listen to families who are concerned about the maternity ward" → 0 (relational register, not agentic)
- "We will ask the government and the new Assembly to continue to improve the social well-being of the people of Martinique" → 0 (demands the government, not the candidate)

Answer 0 otherwise, particularly for:

- proposals stated without any indication of the candidate's action or determination;
- the mere mention of a profession, degrees, or positions held ("engineer, graduate of Centrale"); however, describing a success or an accomplished action counts ("I turned around a struggling company" → 1);
- attacks on opponents that say nothing about the candidate themselves;
- calls to vote, slogans, names of parties or coalitions, and illegible segments.

Ignore grammatical gender: "déterminé" and "déterminée" are transcribed the same way. The text may contain OCR errors: transcribe them as best you can, without correcting them.

Format: Respond only with the number 1 or the number 0, without any other characters.

Table B9: Prompt shared with ChatGPT to classify a sample of paragraphs into agentic language

You are an expert in French political science. You are given a segment (a sentence or a short paragraph) taken from the campaign platform of a candidate in the French legislative elections.

Question: In this segment, does the candidate present themselves in a COMMUNAL manner, focused on others?

The communal register emphasizes connections with others and concern for others: listening, approachability, availability, attention to people, kindness, humility, sincerity, a spirit of dialogue and cooperation, and a willingness to work with residents rather than for them.

Important: Code the MANNER in which the candidate presents themselves, not the TOPIC addressed. The same theme can be addressed in different ways:

- "I will listen to families who are concerned about the future of the maternity ward" → 1
- "Available and close to you, I will hold weekly office hours" → 1
- "I will fight tirelessly to keep the maternity ward open" → 0 (agentic register)
- "The maternity ward must remain open" → 0 (neutral programmatic statement)

Reply with 1 if the segment introduces the candidate (or his team, when he speaks in the first-person plural) in a conversational tone.

Answer 0 otherwise, particularly for:

- proposals stated without any indication of the candidate's relationship with or consideration for people;
- social or solidarity policies described as measures ("increase housing assistance"): this is a topic, not a tone;
- the mere mention of one's profession, degrees, or positions held;
- attacks on opponents, calls to vote, slogans, the names of parties or coalitions ("Ensemble," "Rassemblement national"), and illegible segments.

Ignore the grammatical gender: "attentif" and "attentive" are transcribed the same way. The text may contain OCR errors: transcribe them as best you can, without correcting them.

Format: Respond only with the number 1 or the number 0, without any other characters.

C Empirical strategy

Table C10: Summary statistics

	First round				Second round				Adjustment			
	Women		Men		Women		Men		Women		Men	
	All	Near	All	Near	All	Near	All	Near	All	Near	All	Near
Female-stereotyped	0.114 (0.101)	0.124 (0.092)	0.094 (0.091)	0.103 (0.089)	0.103 (0.087)	0.104 (0.093)	0.072 (0.083)	0.087 (0.090)	-0.025 (0.077)	-0.015 (0.065)	-0.026 (0.083)	-0.015 (0.074)
Male-stereotyped	0.115 (0.136)	0.134 (0.182)	0.104 (0.132)	0.141 (0.192)	0.107 (0.138)	0.128 (0.173)	0.091 (0.127)	0.125 (0.173)	-0.016 (0.085)	-0.012 (0.096)	-0.010 (0.087)	-0.019 (0.108)
Other	0.884 (0.100)	0.883 (0.105)	0.899 (0.103)	0.892 (0.097)	0.896 (0.105)	0.890 (0.115)	0.921 (0.101)	0.906 (0.106)	0.013 (0.081)	0.008 (0.085)	0.018 (0.086)	0.013 (0.092)
Health	0.056 (0.057)	0.059 (0.055)	0.040 (0.055)	0.054 (0.061)	0.052 (0.056)	0.054 (0.057)	0.031 (0.050)	0.049 (0.062)	-0.011 (0.046)	-0.006 (0.041)	-0.008 (0.046)	-0.006 (0.048)
Education	0.065 (0.085)	0.074 (0.069)	0.059 (0.074)	0.055 (0.065)	0.056 (0.062)	0.056 (0.063)	0.043 (0.062)	0.040 (0.055)	-0.015 (0.063)	-0.012 (0.056)	-0.019 (0.068)	-0.013 (0.052)
Justice	0.077 (0.080)	0.088 (0.086)	0.069 (0.083)	0.087 (0.090)	0.085 (0.103)	0.089 (0.104)	0.069 (0.095)	0.082 (0.104)	-0.004 (0.067)	0.000 (0.072)	-0.004 (0.074)	-0.006 (0.075)
Immigration	0.056 (0.122)	0.070 (0.177)	0.049 (0.108)	0.077 (0.181)	0.045 (0.110)	0.058 (0.144)	0.038 (0.094)	0.058 (0.142)	-0.011 (0.111)	-0.020 (0.132)	-0.006 (0.084)	-0.022 (0.150)
Foreign policy	0.041 (0.063)	0.023 (0.039)	0.039 (0.059)	0.018 (0.041)	0.028 (0.047)	0.023 (0.043)	0.023 (0.045)	0.019 (0.037)	-0.007 (0.051)	0.002 (0.035)	-0.012 (0.052)	0.004 (0.035)
Observations	7909	199	19094	223	1674	185	5376	210	1598	173	5151	189

'All' refers to all candidates. 'Near' refers to the candidates where the running variable is within 3 p.p. of the cut-off.

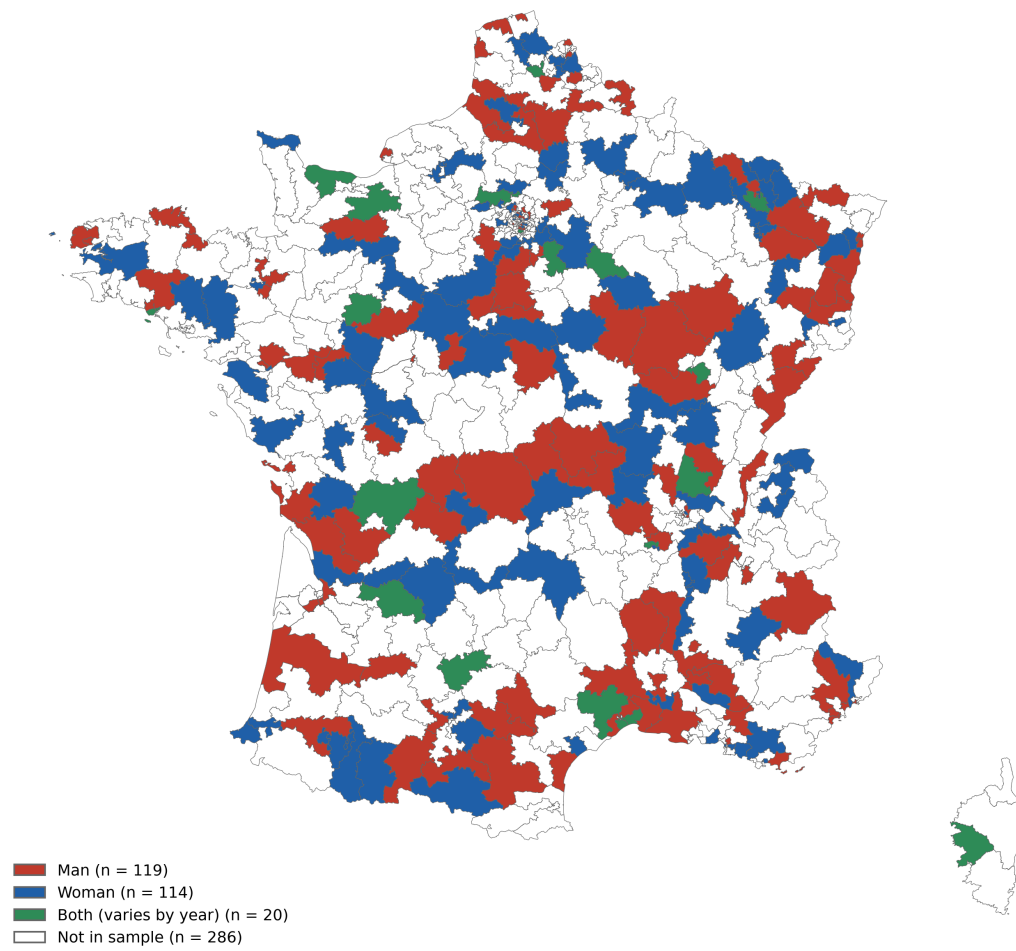
Standard errors are in parenthesis.

Table C11: Regression discontinuity design sampling frame

Year	Districts	Candidates	Mean margin
1981	1	1	7.474
1988	3	3	7.138
1993	138	139	6.276
1997	24	24	4.814
2002	2	2	3.880
2007	2	2	7.777
2012	26	26	5.239
2017	327	352	4.144
2022	326	401	4.756
2024	11	13	4.307

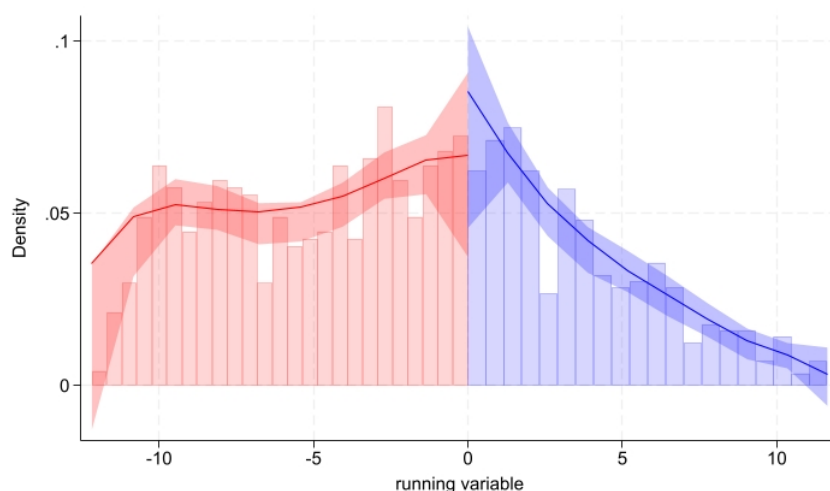
Notes: Column 'Districts' indicates the number of electoral districts included in the regression discontinuity design for the respective year, 'Candidates' refers to the number of candidates included, and 'Mean margin' is the average running variable.

Figure C.3: Geographic variation within 3 percentage points of the cutoff



C.1 Internal validity

Figure C.4: Manipulation testing: Most voted female - most voted male



Notes. The figure shows the density test for races where only one candidate (or none) received 12.5%; the margin is the difference between the most-voted woman and the most-voted man (ignoring the opponent in the second round). The figure illustrates the Cattaneo et al. (2018) manipulation test; p-value 0.600 (do not reject the null hypothesis of no manipulation).

I conduct placebo tests to examine whether there is discontinuity at the threshold for each variable used to predict treatment. I provide information about the construction of each variable. If the information is missing, it is because the name of the dependent variable is self-explanatory.

Turnout: share of enrolled voters that voted. Continuous variable from 0 to 100.

Number candidates: number of candidates running in the electoral district.

Number female candidates: number of female candidates running in the electoral district.

Elected a woman in the past: dummy variable equal to 1 if the district elected a woman in a previous election, 0 otherwise.

Wage gap: Le Barbanchon and Sauvagnat (2022) gender bias estimation; they computed the residualized local earnings gaps after controlling for age, industry, and occupation. The data miss observations for some electoral districts outside of metropolitan France.

Number far-left, left, liberal, right, far-right, other: Number of candidates of the respective ideology.

Sum of far-left, left, liberal, right, far-right, other: sum of the vote share in all candidates of the respective ideology.

Votes: number of votes obtained divided by the number of enrolled voters.

Victory margin: difference between the share of votes obtained in the first round by the candidate and the opponent in the second round.

Run in the past: dummy equal to 1 if the candidate has run for legislative elections in the past, 0 if not.

Out of labor force: dummy equal to 1 if the candidate does not participate in the labor market (e.g., unemployed, retired, student).

High skilled: dummy equal to 1 if the candidate is a high skilled worker (e.g., manager, doctor, university professor, lawyer, journalist), 0 otherwise.

Middle skilled: dummy equal to 1 if it is a middle skilled worker (e.g., technician, travel agent, militar, kindergarten teacher, foreman), 0 otherwise.

Low skilled: dummy equal to 1 if it is a low skilled worker (e.g., minor, craftsman, driver, industrial worker), 0 otherwise.

Far-left, left, liberal, right, far-right: a dummy equal to 1 if the politician is classified as belonging to that ideology, 0 if not.

Run: dummy equal to 1 if the candidate was eligible for the 2nd round and decided to run, 0 if he was eligible but decided not to run.

Available: dummy equal to 1 if the manifesto for the 1st and 2nd round is available, 0 if not.

Table C12: Balancing tests: 1st round electoral district characteristics

	(1) Turnout	(2) Number candidates	(3) Number female candidates	(4) Enrolled voters	(5) Elected a woman in the past	(6) Wage gap
Woman	-0.029 (5.742)	-0.189 (0.716)	0.011 (0.494)	-6,825 (3,379)	0.009 (0.107)	0.003 (0.005)
Observations	963	963	963	963	963	871
Eff. number of obs	326	334	352	390	318	316
Robust p-value	0.930	0.987	0.860	0.041	0.877	0.503
Polyn. order	1	1	1	1	1	1
Bandwidth	2.718	2.824	2.911	3.225	2.619	2.876
Mean, left of threshold	22.359	11.968	4.775	82329	0.241	0.0704

	(7)	(8)	(9)	(10)	(11)	(12)
	Number of far-left	Number of left	Number of liberal	Number of right	Number of far-right	Number of other
Woman	-0.089 (0.232)	-0.144 (0.254)	0.017 (0.089)	-0.003 (0.225)	0.047 (0.200)	-0.177 (0.443)
Observations	963	963	963	963	963	963
Eff. number of obs	365	301	340	453	379	352
Robust p-value	0.909	0.951	0.748	0.952	0.796	0.889
Polyn. order	1	1	1	1	1	1
Bandwidth	3.021	2.473	2.865	3.889	3.107	2.918
Mean, left of threshold	2.026	3.203	0.803	1.829	2.127	1.990

	(13)	(14)	(15)	(16)	(17)	(18)
	Sum of far-left	Sum of left	Sum of liberal	Sum of right	Sum of far-right	Sum of other
Woman	0.204 (0.946)	0.824 (1.440)	0.357 (1.603)	-0.764 (2.164)	-0.623 (1.114)	0.380 (0.657)
Observations	963	963	963	963	963	963
Eff. number of obs	360	355	331	294	334	444
Robust p-value	0.777	0.531	0.684	0.533	0.498	0.704
Polyn. order	1	1	1	1	1	1
Bandwidth	3.000	2.968	2.765	2.329	2.804	3.801
Mean, left of threshold	4.129	12.826	10.210	10.38	10.42	1.325

Table C13: Balancing tests: individual characteristics

	(1)	(2)	(3)	(4)
	Votes	Victory margin	Incumbent	Run in the past
Woman	0.394 (0.490)	0.457 (1.411)	0.019 (0.095)	-0.016 (0.109)
Observations	963	963	963	963
Eff. number of obs	366	338	319	333
Robust p-value	0.546	0.678	0.876	0.883
Polyn. order	1	1	1	1
Bandwidth	3.028	2.849	2.650	2.790
Mean, left of threshold	8.949	-7.377	0.178	0.322

	(5)	(6)	(7)	(8)	(9)
	Occupation				
	Out of labor force	High skilled	Middle skilled	Low skilled	Missing
Woman	0.057 (0.076)	-0.038 (0.094)	-0.026 (0.094)	0.052 (0.047)	0.081 (0.057)
Observations	963	963	963	963	963
Eff. number of obs	334	474	326	246	294
Robust p-value	0.373	0.657	0.712	0.164	0.148
Polyn. order	1	1	1	1	1
Bandwidth	2.802	4.029	2.687	1.946	2.348
Mean, left of threshold	0.080	0.653	0.189	-0.00735	0.0106

	(10)	(11)	(12)	(13)	(14)
	Age group				
	<25	25-44	45-64	>= 65	Missing
Woman	0.015 (0.025)	-0.050 (0.129)	0.114 (0.132)	-0.109 (0.061)	0.022 (0.037)
Observations	963	963	963	963	963
Eff. number of obs	446	280	284	382	381
Robust p-value	0.477	0.571	0.335	0.117	0.564
Polyn. order	1	1	1	1	1
Bandwidth	3.831	2.220	2.246	3.179	3.134
Mean, left of threshold	0.016	0.382	0.459	0.122	0.0228

	(15)	(16)	(17)	(18)	(19)	(20)
	Political ideology					
	Far-left	Left	Liberal	Right	Far-right	Other
Woman	0.036 (0.070)	0.119 (0.116)	0.034 (0.100)	-0.172 (0.080)	0.015 (0.074)	-0.037 (0.035)
Observations	963	963	963	963	963	963
Eff. number of obs	382	322	355	340	365	326
Robust p-value	0.792	0.294	0.795	0.060	0.821	0.330
Polyn. order	1	1	1	1	1	1
Bandwidth	3.168	2.663	2.958	2.861	3.018	2.711
Mean, left of threshold	0.098	0.320	0.230	0.187	0.126	0.0392

Standard errors are in parentheses and clustered at the district-year level. The unit of observation is the candidate. The independent variable is a dummy equal to 1 if the candidate is a woman. Each column reports the results from a separate local polynomial regression. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure.

Table C14: Balancing tests - differences in the ideology of female and male opponents

	(1)	(2)	(3)	(4)	(5)	(6)
	Far-left	Left	Liberal	Right	Far-right	Other
Woman	0.017 (0.023)	-0.106 (0.088)	0.026 (0.149)	0.084 (0.125)	-0.004 (0.104)	0.005 (0.022)
Observations	963	963	963	963	963	963
Eff. number of obs	422	457	304	304	406	373
Robust p-value	0.442	0.230	0.860	0.499	0.967	0.836
Polyn. order	1	1	1	1	1	1
Bandwidth	3.525	3.911	2.496	2.518	3.344	3.091
Mean, left of threshold	0.000	0.210	0.360	0.178	0.228	0.007

Standard errors are in parentheses and clustered at the district-year level. The unit of observation is the candidate. The independent variable is a dummy equal to 1 if the candidate is a woman. Each column reports the results from a separate local polynomial regression. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure.

Table C15: Differences in topic adjustment between right-wing and non-right-wing candidates

	(1)	(2)	(3)
	Female-stereotyped	Male-stereotyped	Other
Woman	0.010 (0.034)	0.010 (0.032)	0.086 (0.047)
Observations	534	534	534
Eff. number of obs	139	169	139
Robust p-value	0.768	0.754	0.066
Polyn. order	1	1	1
Bandwidth	2.167	2.643	2.165
Mean, left of threshold	-0.011	-0.017	-0.040

Standard errors in parentheses are clustered at the district-year level. Each column reports the results from a separate local polynomial regression. The analysis is restricted to males. The variable of interest is a being right-wing rather than another ideology (far-left, left, liberal, far-right or other). Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. The dependent variables take values between 0 and 1, and their computation is similar to the one in Section 3.3.1. The mean gives the average outcome value for non-right-wing candidates at the threshold.

Table C16: Differences in topic adjustment between female and male candidates, by political ideology

Ideology	Female-stereotyped		Male-stereotyped		Other	
	Left	Right	Left	Right	Left	Right
Woman	-0.009 (0.039)	-0.036 (0.031)	0.014 (0.020)	0.081* (0.063)	-0.003 (0.039)	-0.076** (0.056)
Difference p -value	0.500		0.132		0.124	
Robust p -value	0.678	0.108	0.591	0.079	0.685	0.031
Observations	372	591	372	591	372	591
Eff. number of obs.	146	254	129	171	139	177
Polyn. order	1	1	1	1	1	1
Bandwidth	2.942	3.513	2.536	2.417	2.827	2.537
Mean, left of threshold	-0.007	-0.011	-0.012	-0.057	0.008	0.073

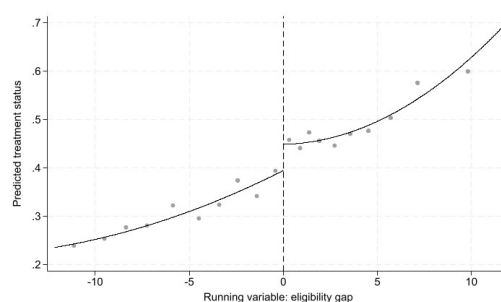
Notes: Standard errors in parentheses are clustered at the district-year level. Each column reports the results from a separate local polynomial regression. The variable of interest is being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. Coefficients are conventional local polynomial estimates; standard errors, confidence intervals, and p -values are robust bias-corrected following Calonico et al. (2014). The dependent variables take values between 0 and 1, and their computation is explained in Section 3.3.1. The mean gives the average outcome value for male candidates at the threshold.

Table C17: Differences in campaigning style (emotion vs reason) between female and male candidates in the first round

	(1)	(2)	(3)	(4)
	All topics	Female-stereotyped	Male-stereotyped	Other
Woman	0.105 (0.177)	-0.067 (0.197)	-0.201 (0.215)	0.097 (0.167)
Observations	963	963	963	963
Eff. number of obs	426	365	293	418
Robust p -value	0.820	0.801	0.258	0.839
Polyn. order	1	1	1	1
Bandwidth	3.561	3.012	2.322	3.479
Mean, left of threshold	-0.302	-0.276	0.176	-0.397

Standard errors in parenthesis are clustered at the district-candidate level. Each column reports the results from a separate local polynomial regression. The variable of interest is a being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. The computation of the dependent variables is explained in Section 3.3.2. The mean gives the average outcome value for male candidates at the threshold.

Figure C.5: General balance test



Notes. Dots represent the local averages of the predicted treatment status (vertical axis). Averages are calculated within quantile bins of the running variable (horizontal axis). The outcome is the candidate's predicted treatment status based on observable characteristics listed in the text. The treatment variable is a dummy equal to 1 if the candidate is a female and qualifies for the second round. The sample is restricted to candidates included in the RDD sample as described in the text. The running variable is the difference between the most-voted woman and the most-voted man (excluding the opponent in the runoff).

Table C18: General balance test

	(1)
Woman	0.040 (0.052)
Observations	963
Eff. number of obs	358
Robust p-value	0.446
Polyn. order	1
Bandwidth	2.998
Mean, left of threshold	0.404

The outcome is the candidate's predicted treatment status based on observable characteristics listed in the text. The outcome is computed as follows: for column (1), first, the treatment variable T is regressed on all 45 baseline variables presented in Tables C12, C13, 1 and C17, and then the treatment status of each candidate is predicted using the regression coefficients. The sample is restricted to candidates included in the RDD sample as described in the text. The independent variable is an indicator equal to 1 if the candidate is a female. I use a nonparametric estimation procedure and MSERD data-driven bandwidths. Standard errors are in parentheses and clustered at the district-year level. The mean gives the average outcome value for the male candidates at the threshold.

Table C19: Differences in sample selection between female and male candidates

	(1)	(2)
	2nd round happens	Platform available
Woman	-0.000 (0.018)	-0.082 (0.089)
Observations	1530	1530
Eff. number of obs	690	563
Robust p-value	0.861	0.368
Polyn. order	1	1
Bandwidth	3.618	2.917
Mean, left of threshold	0.988	0.637

Column (1) is a dummy variable equal to 1 if there is a 2nd round. Column (2) is a dummy variable equal to 1 if the political platform is available for both rounds. The unit of observation is the candidate. The independent variable is a dummy equal to 1 if it is a female candidate. Each column reports the results from a separate local polynomial regression. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. Standard errors are clustered at the district-year level.

D Results

Table D20: Differences between female and male 2nd round political candidates (controlling for enrolled voters)

Panel A: Aggregated topic groups

	(1) Female-stereotyped	(2) Male-stereotyped	(3) Other
Woman	-0.019 (0.015)	0.053 (0.023)	-0.042 (0.023)
Robust p -value	0.263	0.024	0.048
Observations	963	963	963
Eff. number of obs	405	294	330
Polyn. order	1	1	1
Bandwidth	3.338	2.330	2.746
Mean, left of threshold	-0.052	-0.061	0.036

Panel B: Individual topics

	(1) Health	(2) Education	(3) Justice	(4) Immigration	(5) Local	(6) National	(7) Other policy
Woman	-0.023 (0.011)	0.005 (0.014)	0.017 (0.014)	0.056 (0.032)	0.009 (0.018)	-0.064 (0.046)	0.042 (0.049)
Robust p -value	0.029	0.600	0.294	0.058	0.725	0.149	0.307
Observations	963	963	963	963	963	963	963
Eff. number of obs	306	297	390	299	469	383	352
Polyn. order	1	1	1	1	1	1	1
Bandwidth	2.536	2.383	3.219	2.417	4.006	3.189	2.915
Mean, left of threshold	-0.007	-0.052	-0.023	-0.062	0.000	0.207	-0.126

Notes: Each column reports the results from a separate local polynomial regression. Panel A aggregates the topics into female-stereotyped (health and education), male-stereotyped (justice and criminality, and immigration) and remaining topics. Panel B reports the individual topics: health and education compose the female-stereotyped group; justice and criminality, and immigration the male-stereotyped group; and local politics, national politics and other policy (economy, employment and social security, environment, and foreign policy) the remaining topics. The variable of interest is being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. Coefficients are conventional local polynomial estimates. Conventional standard errors, clustered at the district-year level, are in parentheses; p -values are robust bias-corrected following Calonico et al. (2014). The dependent variables take values between 0 and 1, and their computation is explained in Section 3.3.1. The mean gives the average outcome value for male candidates at the threshold.

Table D21: Differences between female and male 2nd round political candidates (covariate-adjusted form)

Panel A: Aggregated topic groups

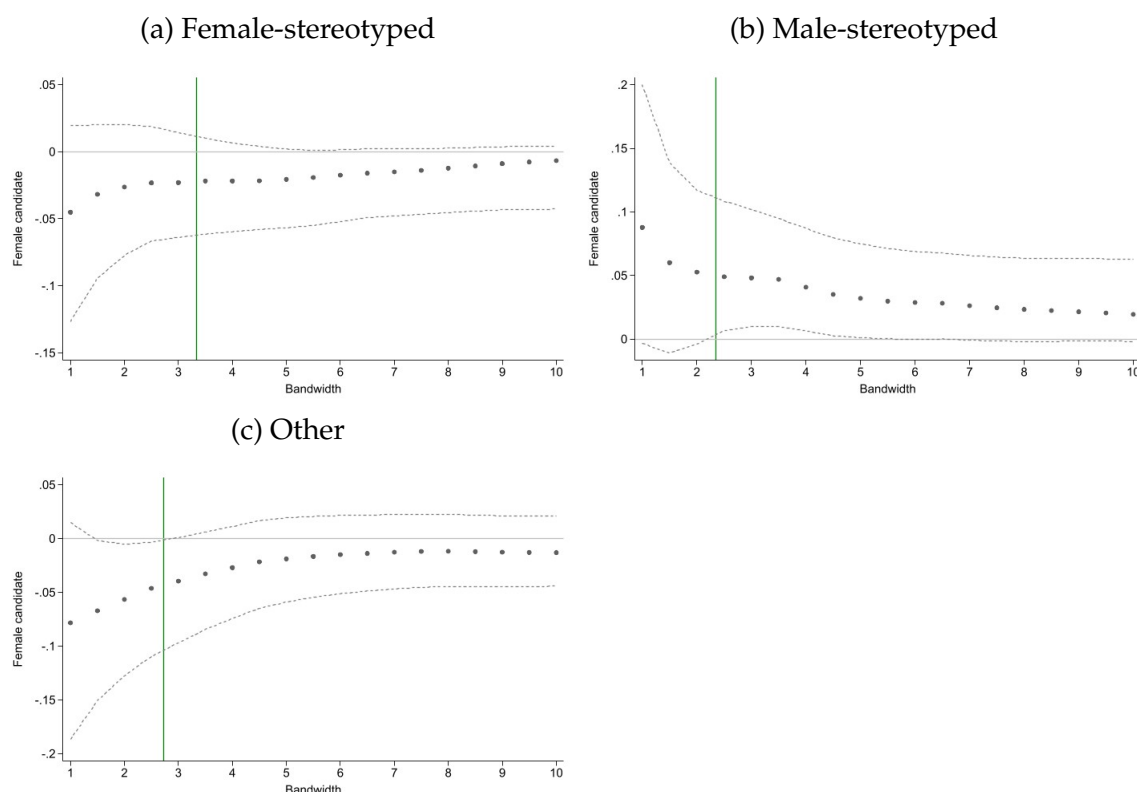
	(1) Female-stereotyped	(2) Male-stereotyped	(3) Other
Woman	-0.020 (0.016)	0.050 (0.020)	-0.041 (0.019)
Robust p -value	0.261	0.019	0.033
Observations	963	963	963
Eff. number of obs	300	295	369
Polyn. order	1	1	1
Bandwidth	2.462	2.367	3.061
Mean, left of threshold	-0.015	0.008	0.015

Panel B: Individual topics

	(1) Health	(2) Education	(3) Justice	(4) Immigration	(5) Local	(6) National	(7) Other policy
Woman	-0.022 (0.010)	0.005 (0.013)	0.018 (0.012)	0.043 (0.026)	-0.008 (0.019)	-0.072 (0.044)	0.058 (0.049)
Robust p -value	0.018	0.703	0.217	0.122	0.614	0.081	0.206
Observations	963	963	963	963	963	963	963
Eff. number of obs	352	290	366	287	334	288	297
Polyn. order	1	1	1	1	1	1	1
Bandwidth	2.919	2.308	3.038	2.257	2.803	2.292	2.382
Mean, left of threshold	0.003	-0.019	-0.103	0.127	0.137	-0.337	0.016

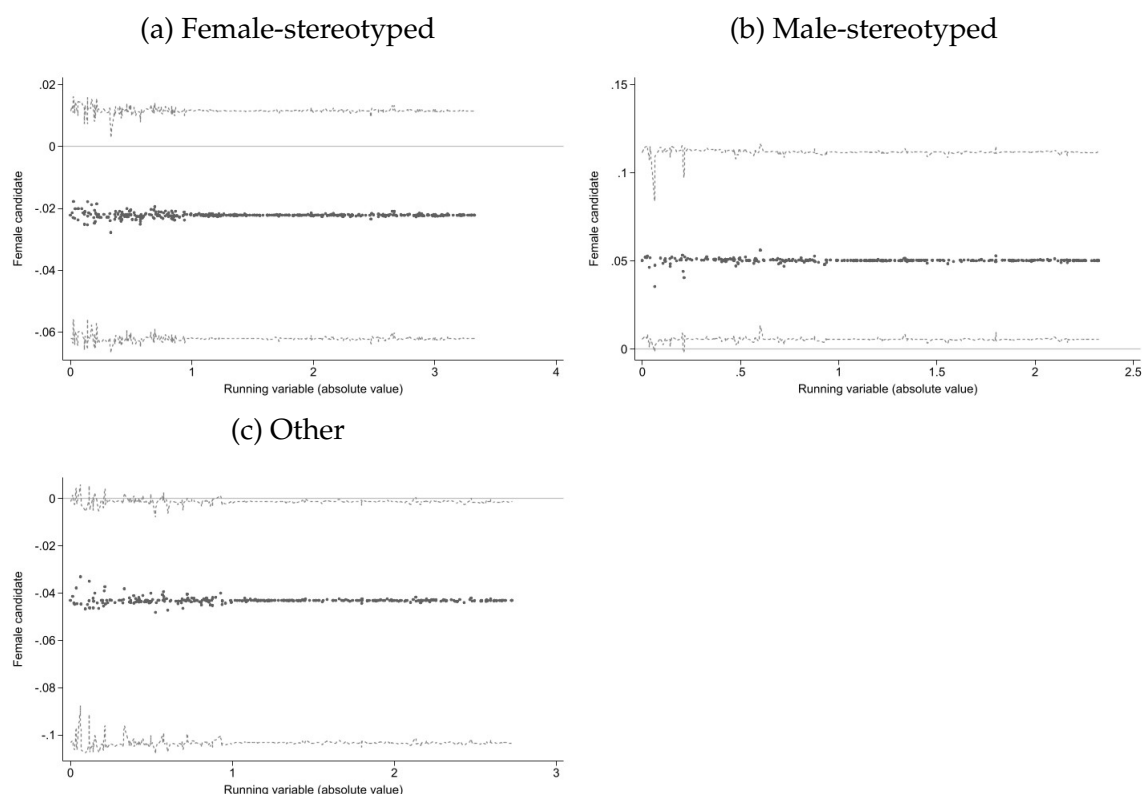
Notes: Each column reports the results from a separate local polynomial regression. Panel A aggregates the topics into female-stereotyped, male-stereotyped, and remaining topics. Panel B reports the individual topics: health and education compose the female-stereotyped group; justice and criminality, and immigration the male-stereotyped group; and local politics, national politics and other policy (economy, employment and social security, environment, and foreign policy) the remaining topics. The variable of interest is being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. Coefficients are conventional local polynomial estimates. Conventional standard errors, clustered at the district-year level, are in parentheses; p -values are robust bias-corrected following Calonico et al. (2014). The dependent variables take values between 0 and 1, and their computation is explained in Section 3.3.1. The mean gives the average outcome value for male candidates at the threshold. All regressions include the covariates listed in Tables C12 and C13.

Figure D.6: Gender differences in topics adjustment between 1st and 2nd round - Robustness to bandwidth choice



Notes. These figures show the sensitivity of the point estimate to bandwidth choice. Dots represent the estimated treatment effect using different bandwidths (horizontal axis). Dotted lines represent the 95-percent robust confidence interval. The estimates are reported for values of the bandwidth from 1 to 10 percentage points, in steps of 0.5 percentage points. The vertical green line gives the value of the MSERD optimal bandwidth used in the main estimation. The outcome is the change in the respective topic coverage between the first and the second round. The independent variable is an indicator equal to 1 if the female candidate reached the second round, 0 if it is a male.

Figure D.7: Gender differences in topics adjustment between 1st and 2nd round - Leave one candidate out

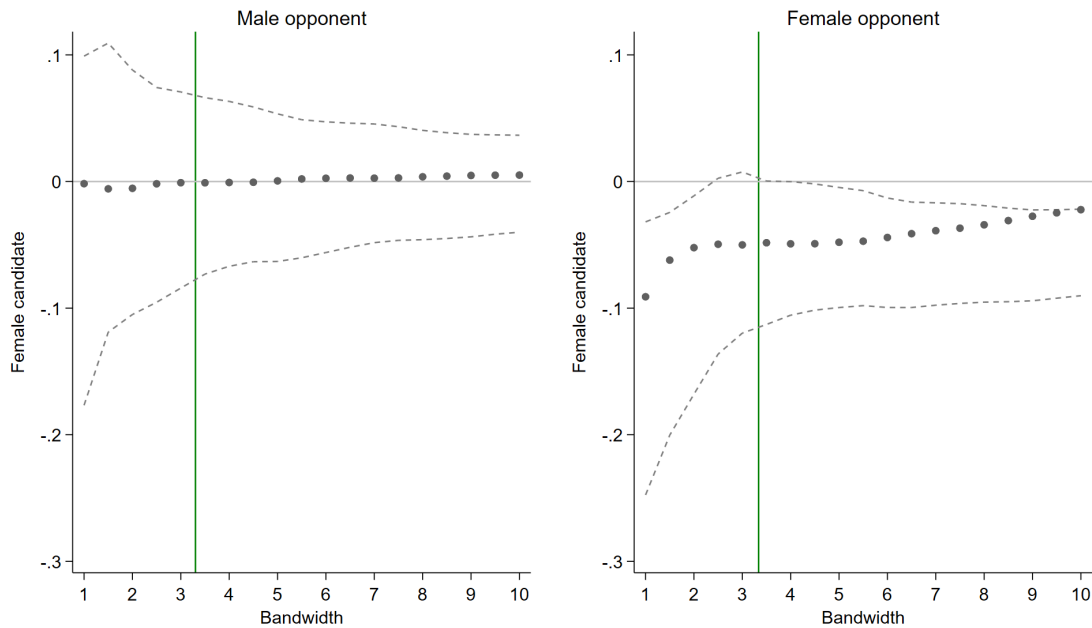


Notes. These figures show the sensitivity of the point estimate when removing one candidate at time. The very first dot on each graph provides the baseline estimate. Then, each dot provides the point estimate associated with removing a given candidate. Candidates are ordered based on the absolute value of their running variable, so that the second dot provides the point estimate associated with removing the candidate the closest to the cutoff. Dotted lines represent the 95-percent robust confidence interval. The outcome is the change in the respective topic coverage between the first and the second round. The independent variable is an indicator equal to 1 if the female candidate reached the second round, 0 if it is a male.

E Mechanisms

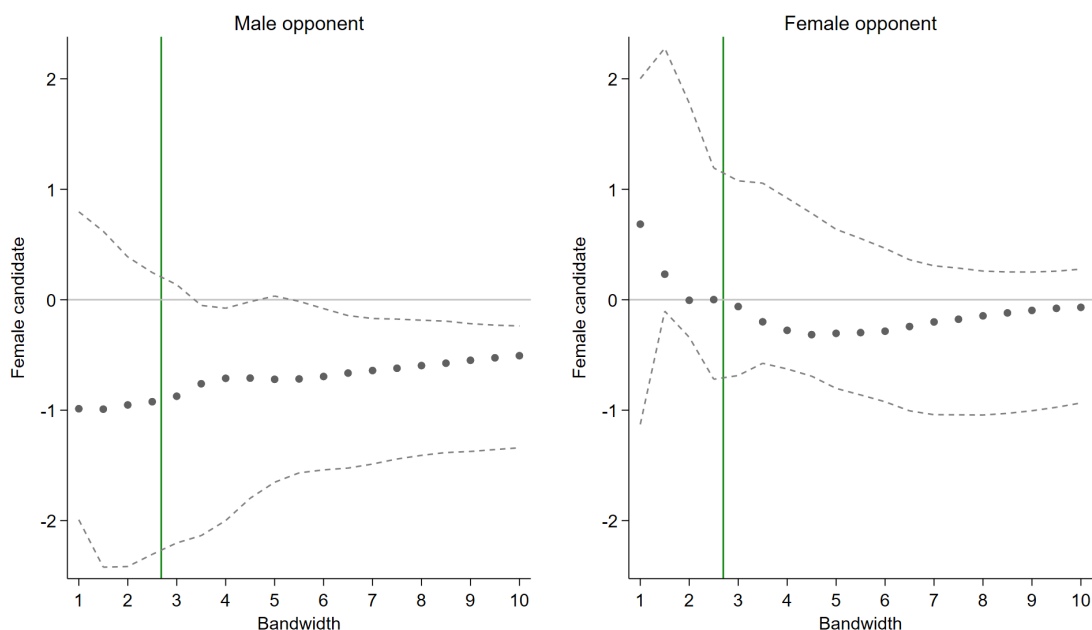
E.1 Do candidates strategically adapt to the gender of their opponent?

Figure E.8: Differences in female-stereotyped topic adjustment between female and male candidates depending on the gender of the 2nd round opponent - Robustness to bandwidth choice



Notes. These figures show the sensitivity of the point estimate to bandwidth choice. Dots represent the estimated treatment effect using different bandwidths (horizontal axis). Dotted lines represent the 95-percent robust confidence interval. The estimates are reported for values of the bandwidth from 1 to 10 percentage points, in steps of 0.5 percentage points. The vertical green line gives the value of the MSERD optimal bandwidth used in the main estimation. The outcome is the change in the female-stereotyped topic coverage between the first and the second round. The independent variable is an indicator equal to 1 if the female candidate reached the second round, 0 if it is a male.

Figure E.9: Differences in agentic-communal adjustment between female and male candidates depending on the gender of the 2nd round opponent - Robustness to bandwidth choice



Notes. These figures show the sensitivity of the point estimate to bandwidth choice. Dots represent the estimated treatment effect using different bandwidths (horizontal axis). Dotted lines represent the 95-percent robust confidence interval. The estimates are reported for values of the bandwidth from 1 to 10 percentage points, in steps of 0.5 percentage points. The vertical green line gives the value of the MSERD optimal bandwidth used in the main estimation. The outcome is the change in the agentic-communal language between the first and the second round. The independent variable is an indicator equal to 1 if the female candidate reached the second round, 0 if it is a male.

E.2 Gender differences and electoral competitiveness

Table E22: Differences in topics adjustment between female and male 2nd round political candidates - Heterogeneity by victory margin in the 1st round between the candidate and the opponent by topic

	Health		Education		Justice		Immigration		Foreign policy	
	Below	Above	Below	Above	Below	Above	Below	Above	Below	Above
Panel A: Heterogeneity with discretized victory margin										
Woman	-0.043 (0.018)	-0.012 (0.029)	0.011 (0.028)	-0.003 (0.030)	0.014 (0.027)	0.018 (0.027)	0.077 (0.083)	0.029 (0.046)	0.000 (0.017)	0.003 (0.022)
P-value (above vs. below)	0.134		0.372		0.625		0.238		0.761	
Observations	481	482	481	482	481	482	481	482	481	482
Eff. number of obs	134	169	159	163	234	181	156	207	214	145
Robust p-value	0.003	0.955	0.602	0.462	0.465	0.964	0.102	0.604	0.350	0.735
Polyn. order	1	1	1	1	1	1	1	1	1	1
Bandwidth	1.944	3.022	2.521	2.900	3.926	3.167	2.444	3.620	3.455	2.531
Mean, left of threshold	0.001	0.009	-0.026	-0.009	-0.010	-0.013	-0.072	-0.035	0.001	0.000
Woman at zero margin	-0.007 (0.026)		-0.006 (0.031)		-0.001 (0.028)		-0.010 (0.053)		0.003 (0.021)	
Robust p-value	0.748		0.434		0.386		0.139		0.553	
Woman victory margin	0.003 (0.002)		-0.001 (0.003)		-0.003 (0.004)		-0.010 (0.008)		-0.000 (0.002)	
Robust p-value	0.049		0.353		0.174		0.007		0.818	
Polyn. order	1		1		1		1		1	
Bandwidth	2.476		2.425		3.293		2.432		2.632	
Eff. number of obs	604		598		798		600		636	
Observations	963		963		963		963		963	

Victory margin is the difference between the share of votes the candidate received in the first round and the opponent's share in the second-round. In Panel A, Below refers to candidates whose victory margin is below the median, while Above refers to candidates whose victory margin is above the median. In Panel B, victory margin is a continuous variable. Standard errors in parentheses are clustered at the district-year level. Each column reports the results from a separate local polynomial regression. The variable of interest is being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. The heterogeneity analysis is performed following Calonico et al. (2025). The dependent variables take values between 0 and 1, and their computation is explained in Section 3.3.1. The mean gives the average outcome value for male candidates at the threshold.

Table E23: Differences in campaign funding between female and male 2nd round political candidates

	(1) Total expenditures	(2) Total revenues	(3) Personal contribution	(4) Party contribution	(5) Private donation
Woman	870 (18,285)	1,740 (18,658)	-249 (7,042)	-69 (6,998)	2,947 (12,746)
Robust <i>p</i> -value	0.906	0.867	0.993	0.865	0.712
Observations	915	915	914	914	914
Eff. number of obs	366	370	344	383	397
Polyn. order	1	1	1	1	1
Bandwidth	3.104	3.143	2.943	3.237	3.384
Mean, left of threshold	48,632	49,579	28,883	6,454	12,328

Notes: Standard errors in parentheses are clustered at the district-year level. Each column reports the results from a separate local polynomial regression. The variable of interest is being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. Coefficients are conventional local polynomial estimates; standard errors, confidence intervals, and *p*-values are robust bias-corrected following Calonico et al. (2014). The dependent variables are, in euros: total amount raised by the candidate during the campaign, total expenditures, own funds, contributions from the party, and private donations. The sample is restricted to the candidates used in the main analysis. The mean gives the average outcome value for male candidates at the threshold.

E.3 Discussion of alternative mechanisms

Table E24: Differences in topics adjustment between female and male 2nd round political candidates

	Female-stereotyped	Male-stereotyped	Other
Woman	-0.034 (0.026)	0.027 (0.031)	-0.013 (0.033)
Robust p-value	0.334	0.490	0.535
Woman $\times$ Elim. women	-0.004 (0.005)	0.007 (0.009)	0.000 (0.007)
Robust p-value	0.190	0.237	0.485
Observations	963	963	963
Eff. number of obs	405	294	329
Polyn. order	1	1	1
Bandwidth	3.331	2.349	2.732

'Elim. women' stands for the share of votes obtained by the eliminated women in the first round demeaned by election-year. Standard errors in parentheses are clustered at the district-year level. Each column reports the results from a separate local polynomial regression. The variable of interest is being a woman rather than a man. Separate polynomials are fitted on each side of the threshold. The polynomial order is 1, and the optimal bandwidths are derived under the MSERD procedure. The heterogeneity analysis is performed following Calonico et al. (2025). The dependent variables take values between 0 and 1, and their computation is explained in Section 3.3.1. The mean gives the average outcome value for male candidates at the threshold.